



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

Lecture 1: CA6121 AI in Project Management

Introduction and Overview

Presented by Dr Ravish Maqsood



Prepared by: Ravish Maqsood

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OBJECTIVES

- Introduction of basic concepts in project management with coverage of evolving trends and application such as sustainability, artificial intelligence.
- The learning helps to apply the Artificial Intelligence (AI) concepts in executing projects which includes design, procurement, and execution phases.
- This equips the participants to enhance their skill set, improve employability, and career growth.



Why should I take this course?

Role of AI in project management is growing

- Automated task management
- Predictive analytics
- Enhanced collaboration
- Resource allocation
- Risk management
- Better decision making
- Many more...

Improving skills

Employability

Future trends

Improving Efficiency

Improving Productivity

Learning

Staying ahead



OBJECTIVES

- Key concepts covered:
 - project integration
 - scope management
 - project planning
 - project budget & cost management
 - procurement
 - project risks management
 - AI tools & applications



Course Learning Outcomes (ILOs)

1. Familiarize and understand, project life cycle, importance and relevance of project management concepts.
2. Understand and apply evolving technological advancement applications with Artificial Intelligence (AI) concepts in project management.
3. Communicate and better engagement with internal and external stakeholders.
4. Plan and schedule activities and resources for a project.
5. Apply tools and knowledge to improve project efficiency and productivity.



Learning and Teaching Approach

Approach	How does this approach support you in achieving the learning outcomes?
Online Recorded Lectures	Pre-recorded online lectures on the topics are provided for students to go through at their own convenience prior coming to the tutorial classes.
In class tutorials using flip class methodology wherever possible	Tutorials shall be held face to face and will have in-class discussions with hands-on exercises. This helps the students to understand the concepts mentioned in the lectures. Flip classroom approach and discussions helps better engagement and faster learning of the concepts.
Online Consultation	Group project consultations and Q&A session (each group is assigned a time slot)



Assessment Structure

No.	Component	Weightage	Team / Individual	Remarks
CA1	Continuous Assessment (CA1): Class Attendance and Exercises	20%	Individual	Students are required to attend all the classes in person to secure your attendance and participation marks.
CA2	Continuous Assessment (CA2): Final Quiz (Entire syllabus)	30%	Individual	Students are required to attend final quiz in-person.
CA3	Continuous Assessment (CA3): Group Project Report	30%	Team	Group project is a teamwork and all are required to participate.
CA4	Continuous Assessment (CA4): Final Presentation	20%	Team	Group project presentation is a teamwork and all are required to participate. Every student is expected to present at least some portion of the group work.



Reading and References

1. A guide to the Project Management body of Knowledge “**PMBOK guide eight edition**”, by Project Management Institute (PMI), Nov 2025 (expected).
2. A guide to the Project Management body of Knowledge “PMBOK guide sixth edition”, by Project Management Institute (PMI), Sep 2017.
3. A guide to the Project Management body of Knowledge “PMBOK guide seventh edition”, by Project Management Institute (PMI), August 2021.
4. Project Management for Practice, a guide and toolbox for successful projects, by Daud Alam ; Uwe Guhl, Springer,2022; second edition
5. Agile Product and Project Management: A Step-by-Step Guide to Building the Right Products Right, by Mariya Breyter, Apress,2022
6. Introduction to Project Management: A Source Book for Traditional PM Basics, by Davies A. Igberaese, Routledge, 2023

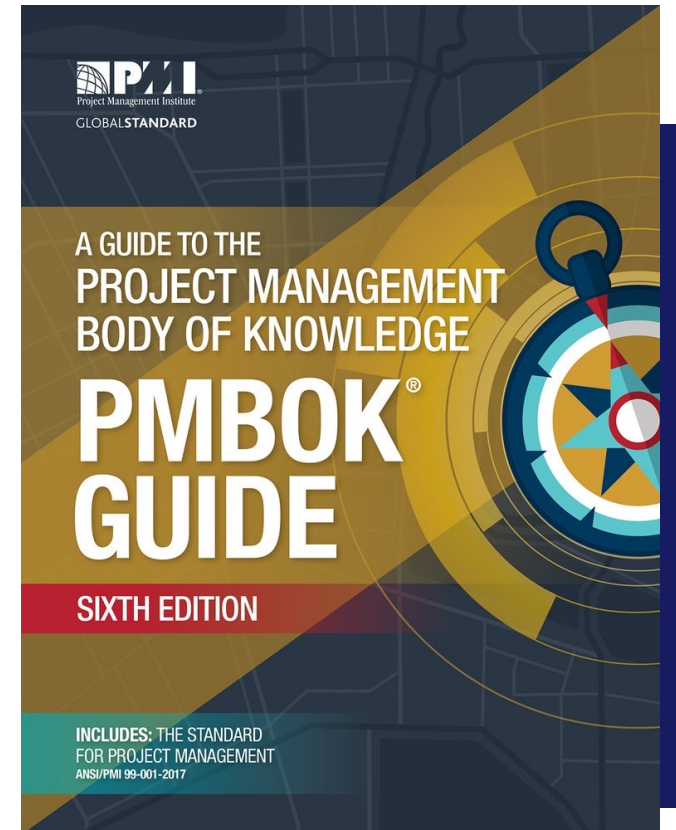


Reading and References

A guide to the Project Management body of Knowledge

“PMBOK guide sixth edition”

by Project Management Institute (PMI), Sep 2017



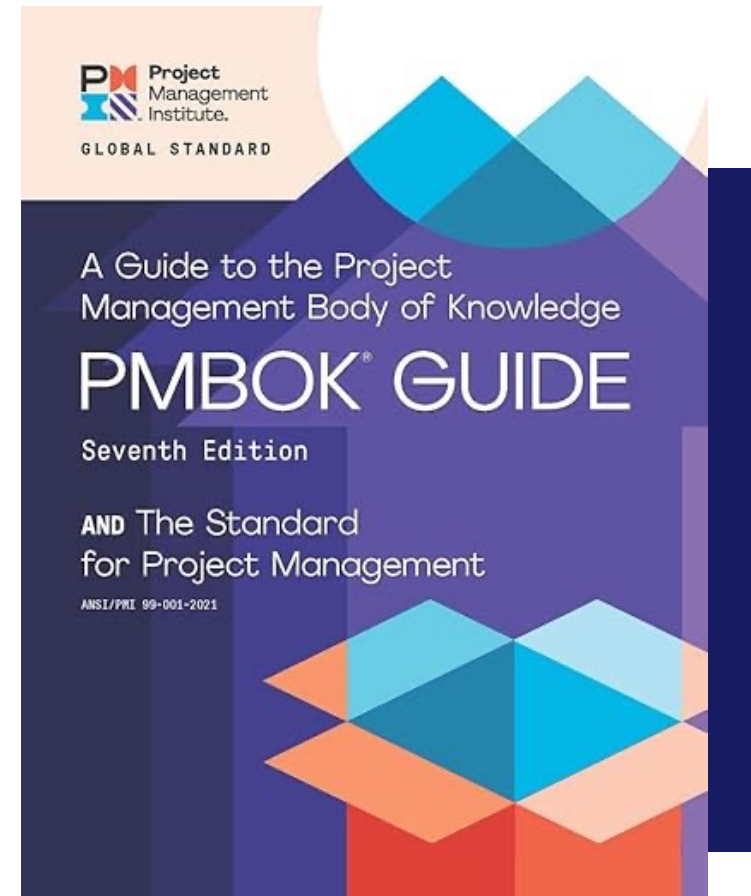


Reading and References

A guide to the Project Management body of Knowledge

“PMBOK guide seventh edition”

by Project Management Institute (PMI), August 2021





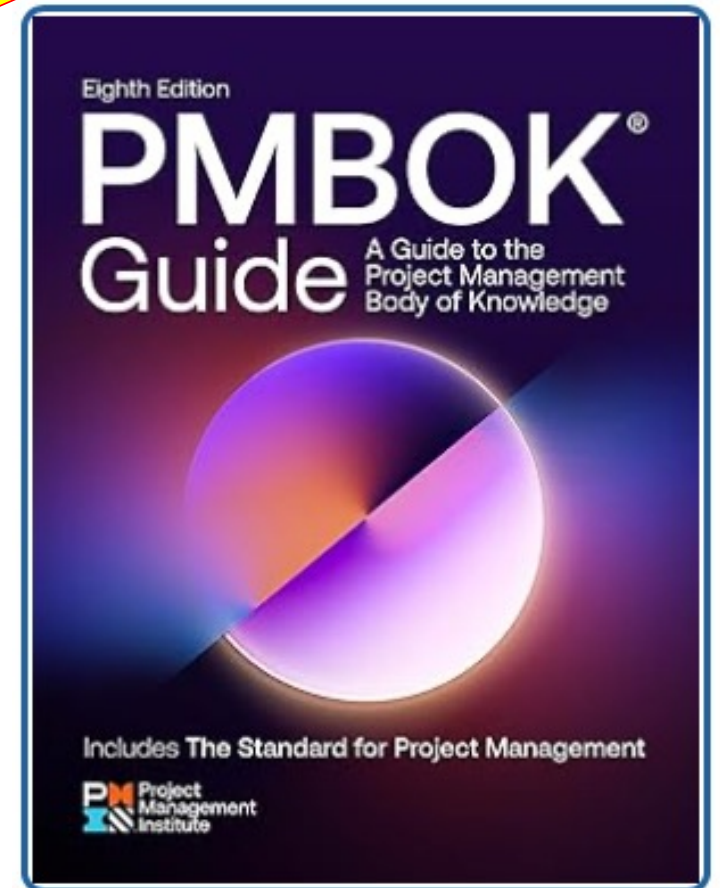
Reading and References

A guide to the Project Management body of Knowledge

“PMBOK guide eighth edition”

by Project Management Institute (PMI), Nov 2025

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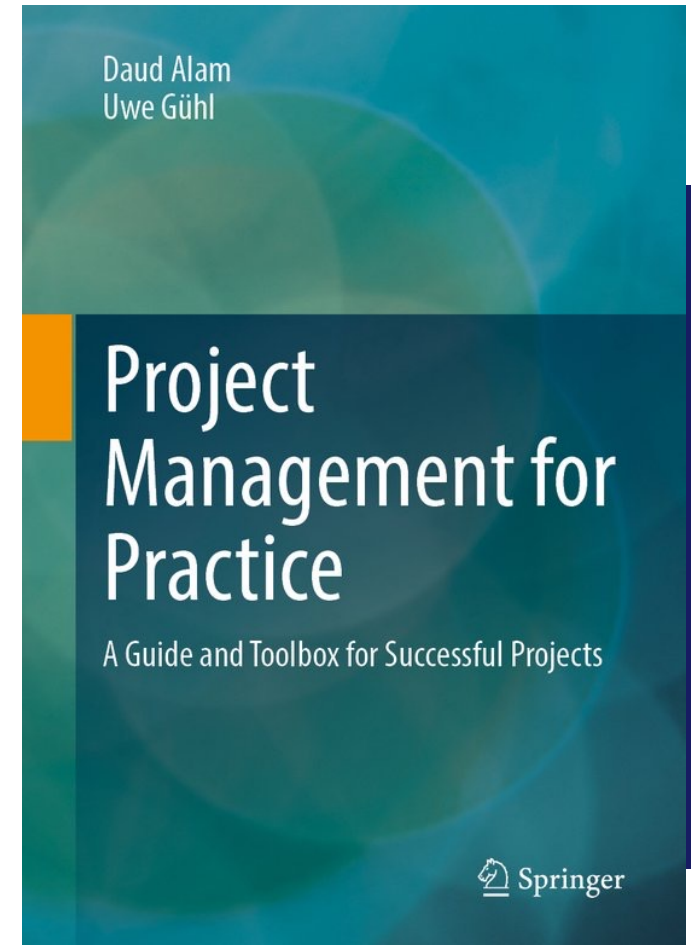




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Project Management for Practice, a guide and toolbox for successful projects

by Daud Alam ; Uwe Guhl, Springer, 2022; second edition

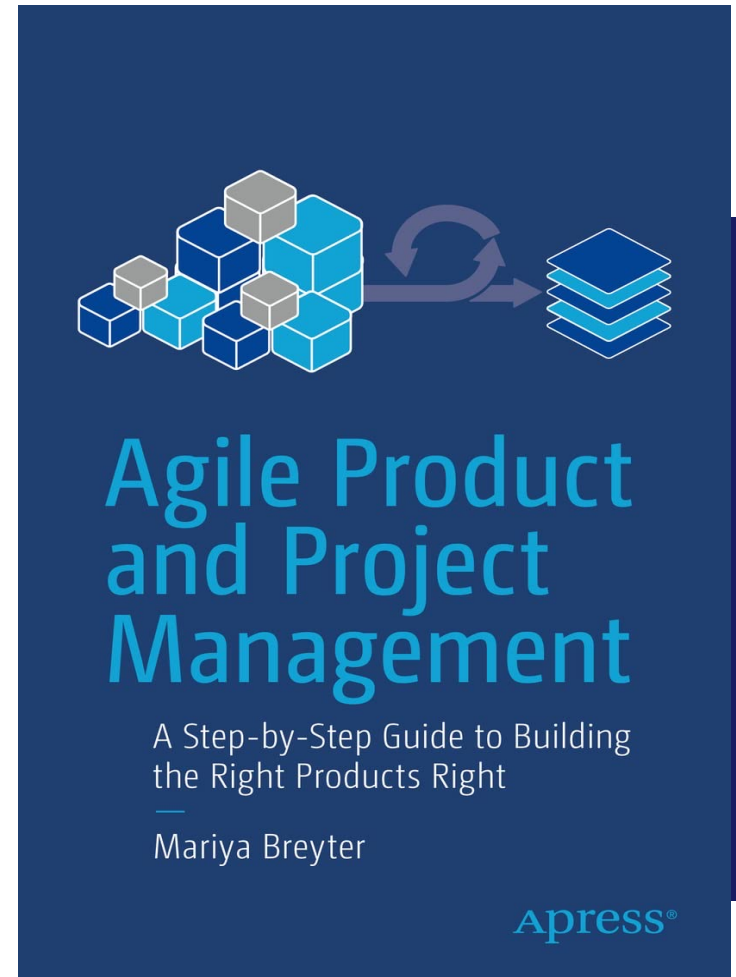




Reading and References

Agile Product and Project Management: A Step-by-Step Guide to Building the Right Products Right

by Mariya Breyter, Apress, 2022

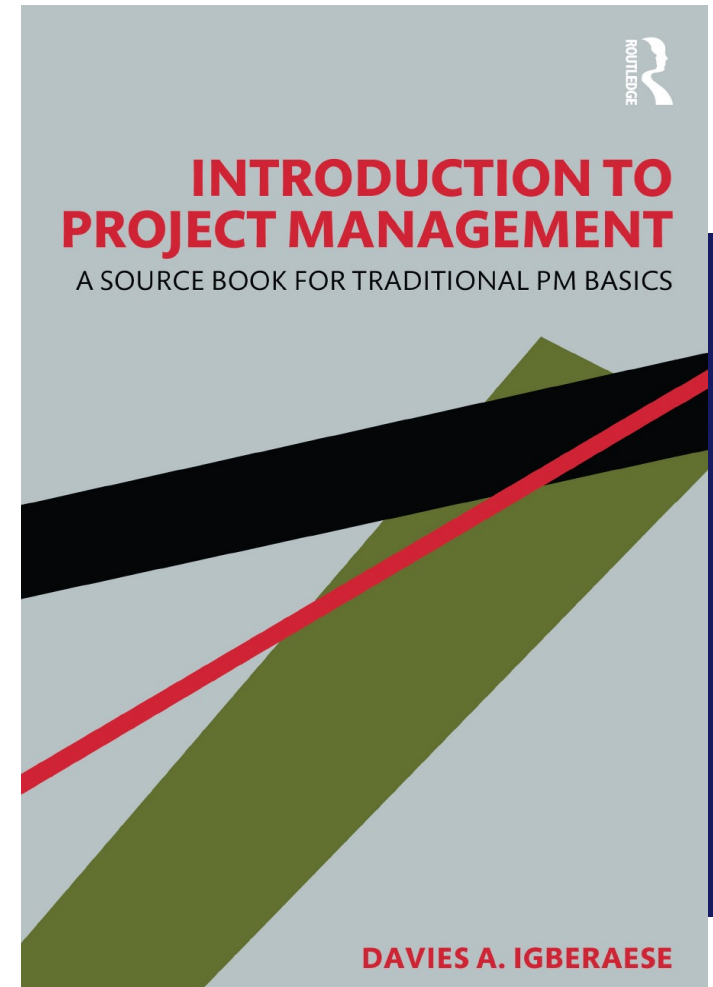




Reading and References

Introduction to Project Management: A Source Book for Traditional PM Basics

by Davies A. Igberaese, Routledge, 2023





Why am I here?

- Facilitate learning process
- Help you to understand the concepts of Project Management
 - In turn you will be able to have good understanding
 - Help in planning for your career path
 - Learning concepts of project management may help you in day-to-day life while planning your activities.
- Learn more by sharing my experience with you.





Project Management: Introduction

Understanding project management



What is a Project?

PMBOK 6th Edition

A project is a temporary endeavor undertaken to create a unique product, service, or result.

Source PMBOK 6th Edition



What is a Project?

PMBOK 6th Edition

Projects are:

- Unique
- Temporary endeavour
- Drive change
- Enable business value creation

Source PMBOK 6th Edition



Why there is a need for a project?

- Market Demand
- Customer Request
- Technological Advances
- Environmental Considerations
- Legal Requirements
- Social Needs



Projects must be aligned with the strategic plan of the organization.



What is a Project Management?

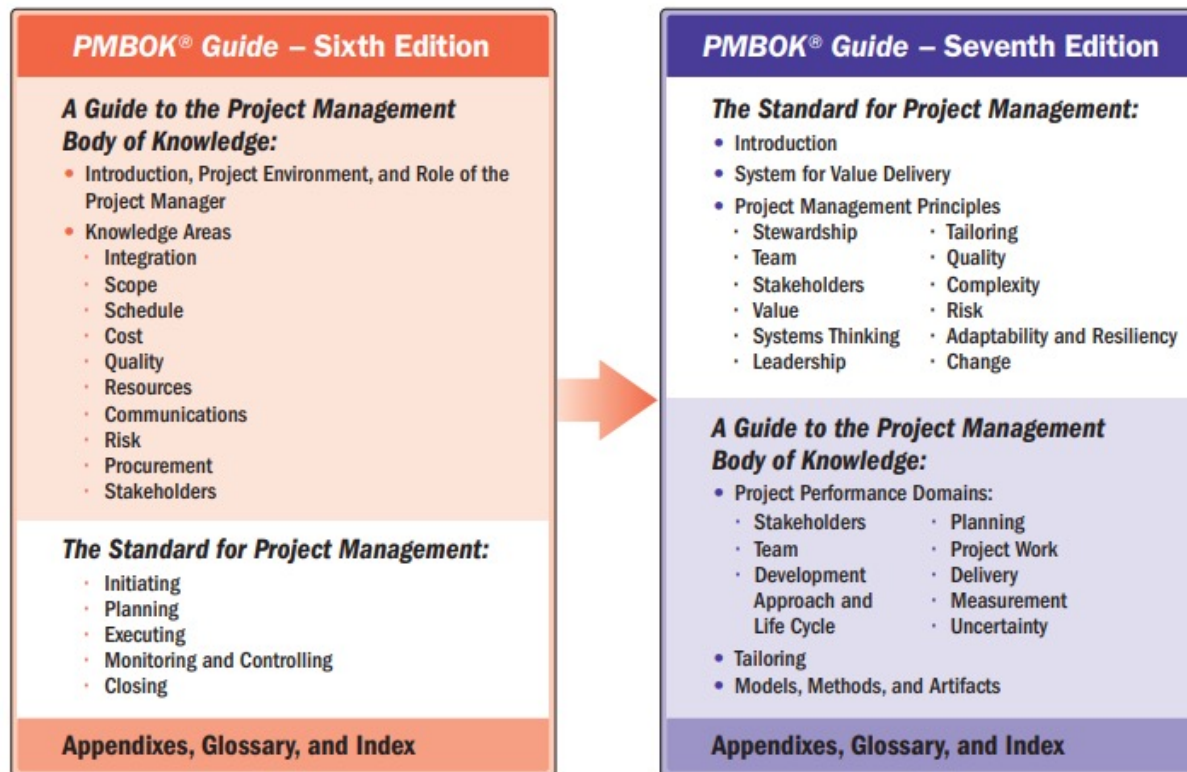
- Project Management is the application of **knowledge, skills, tools and techniques** to meet project requirements.
- It is accomplished using processes:
 1. Initiating
 2. Planning
 3. Executing
 4. Monitoring and Controlling
 5. Closing

Source PMBOK 6th Edition



Project Management

PMBOK 6th Edition to 7th Edition





Project Management

PMBOK 6th Edition

Guide to PM Body of Knowledge 10 Knowledge Areas:

1. Project **Integration** Management
2. Project **Scope** Management
3. Project **Time** Management
4. Project **Cost** Management
5. Project **Quality** Management
6. Project **Human Resource** Management
7. Project **Communications** Management
8. Project **Risk Management**
9. Project **Procurement** Management
10. Project **Stake Holder** Management

PMBOK 7th Edition

Standards (Processes) of Project Management

- Systems for Value Delivery
- **Project Management Principles:**
 1. Stewardship
 2. Team
 3. Stakeholders
 4. Value
 5. Systems Thinking
 6. Leadership
 7. Tailoring
 8. Quality
 9. Complexity
 10. Risk
 11. Adaptability and Resiliency
 12. Change



Project Management

PMBOK 8th Edition

PMBOK 6th Edition

5 Standards (Processes) of Project Management

1. Project Initiation
2. Project Planning
3. Project Execution
4. Project Monitoring and Control
5. Project Closeout

PMBOK 7th Edition

Guide to PM Body of Knowledge

- Systems for Value Delivery
- **Project Performance Domains:**
 1. Stakeholders
 2. Team
 3. Development Approach and Life Cycle
 4. Planning
 5. Project Work
 6. Delivery
 7. Measurement
 8. Uncertainty



Project Management

PMBOK 8th Edition

- Reverting to approach like 6th Edition
- This edition presents five Project Management Focus Areas: [Initiating](#), [Planning](#), [Executing](#), [Monitoring and Controlling](#), and [Closing](#).
- Sustainability

What's new in the **Eighth Edition**:

- Updated definitions of foundational terms and concepts, aligned to current global practice and language accessibility.
- A simplified set of six actionable project management principles.
- The re-imagining of Process Groups as five practical Focus Areas.
- Integration of technical ways of working and other core concepts into seven comprehensive Project Performance Domains, including 40 newly evolved processes



Project Management: Six Principles (PMBOK 8th Addition)

Collectively, these principles describe the mindset of project management, which in turn guides the mechanics of project management.

1. Adopt a Holistic View.
2. Focus on Value.
3. Embed Quality Into Processes and Deliverables.
4. Be an Accountable Leader.
5. Integrate Sustainability Within All Project Areas.
6. Build an Empowered Culture.



Project Management: Six Principles (PMBOK 8th Addition)

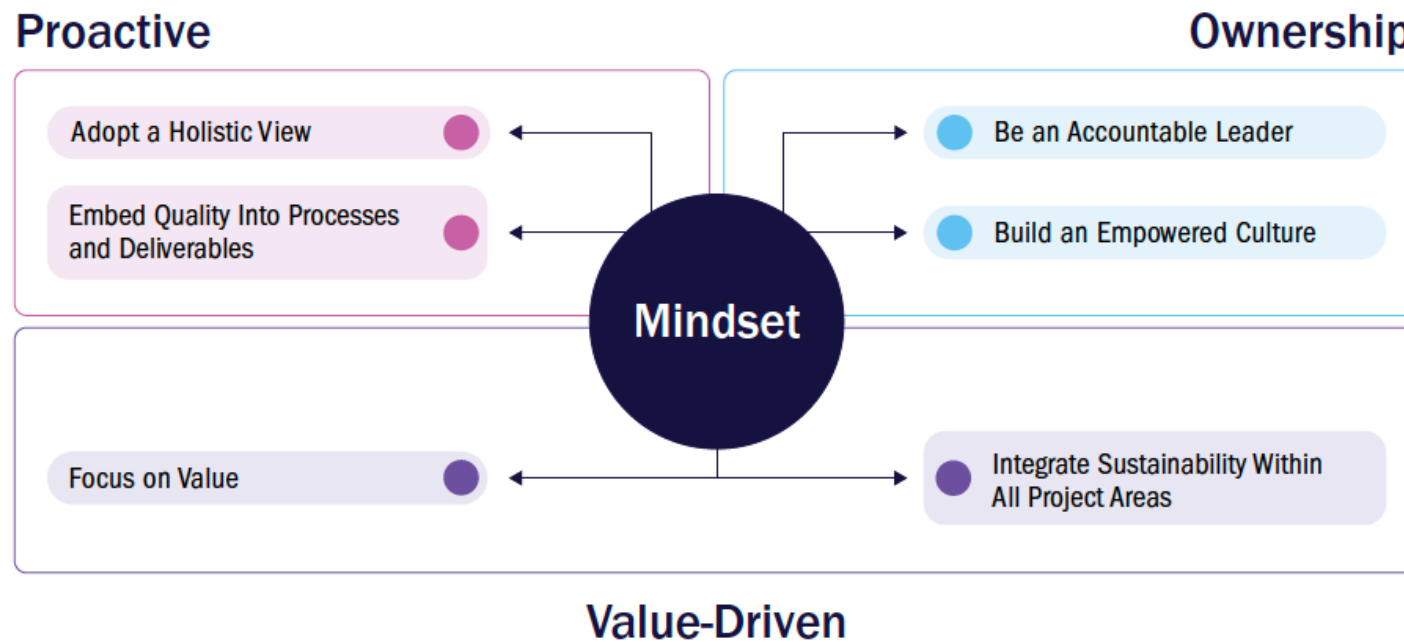


Figure 3-1. The Project Management Mindset



Project Management

PMBOK 8th Edition

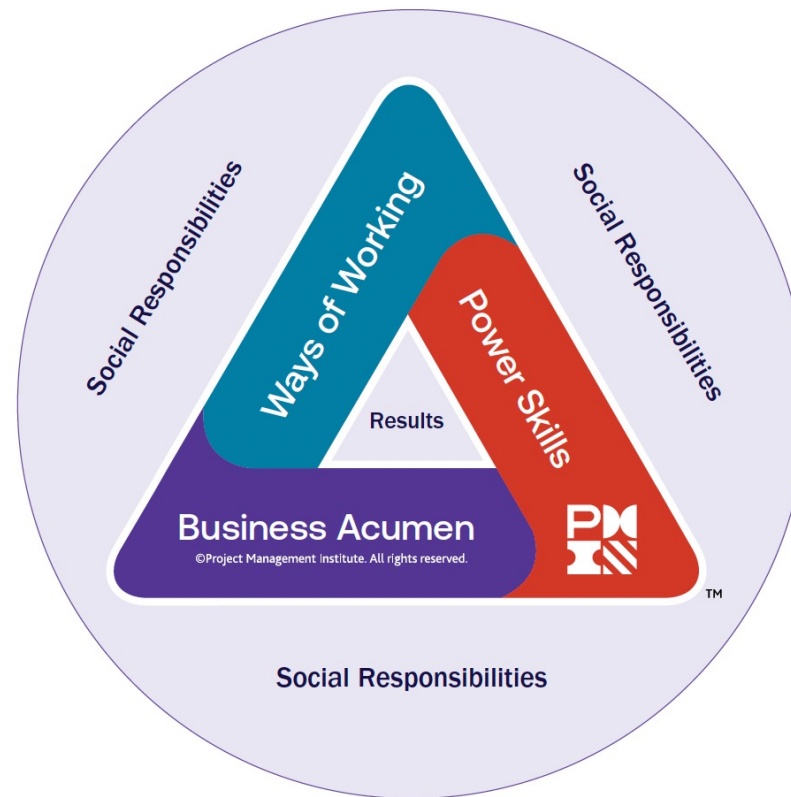


Figure 2-7. Project Management Team Competencies

Source: PMBOK 8th Edition





Project Management

PMBOK 8th Edition

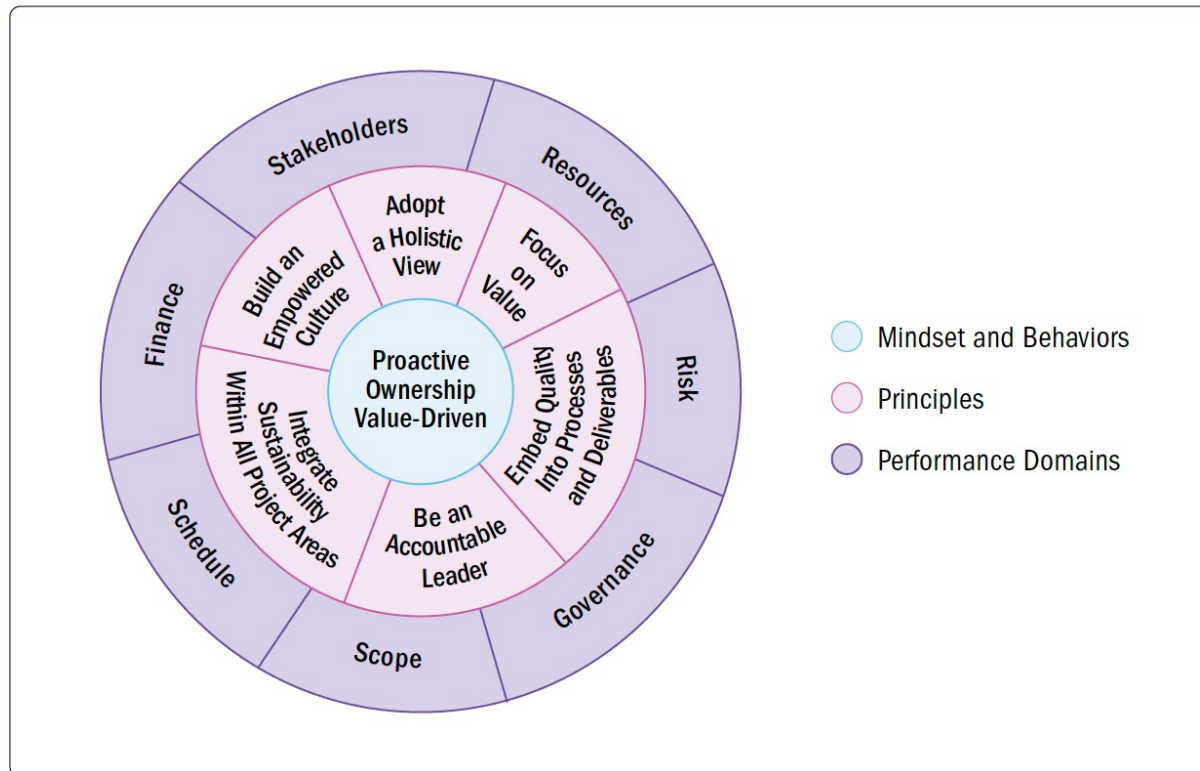


Figure 1-1. Relationship Between Project Management Principles and Project Management Performance Domains

Source: PMBOK 8th Edition





Project Management

PMBOK 8th Edition

Table 2-1. Mapping Project Management Focus Areas to Performance Domains

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Governance	Initiate Project or Phase	- Integrate and Align Project Plans - Plan Sourcing Strategy	- Manage Project Execution - Manage Quality Assurance - Manage Project Knowledge	- Monitor and Control Project Performance - Assess and Implement Changes	Close Project or Phase
Scope		- Plan Scope Management - Elicit and Analyze Requirements - Define Scope - Develop Scope Structure		- Monitor and Control Scope - Validate Scope	
Schedule		- Plan Schedule Management - Develop Schedule		Monitor and Control Schedule	
Finance		- Plan Financial Management - Estimate Costs - Develop Budget		Monitor and Control Finances	

Table 2-1. Mapping Project Management Focus Areas to Performance Domains

Performance Domains	Project Management Focus Areas				
	Initiating Focus Area	Planning Focus Area	Executing Focus Area	Monitoring and Controlling Focus Area	Closing Focus Area
Stakeholders	Identify Stakeholders	- Plan Stakeholder Engagement - Plan Communications Management	- Manage Stakeholder Engagement - Manage Communications	- Monitor Stakeholder Engagement - Monitor Communications	
Resources		- Plan Resource Management - Estimate Resources	- Acquire Resources - Lead the Team	Monitor and Control Resourcing	
Risk		- Plan Risk Management - Identify Risks - Perform Risk Analysis - Plan Risk Responses	Implement Risk Responses	Monitor Risks	

Source: PMBOK 8th Edition





Project Management

PMBOK 8th Edition

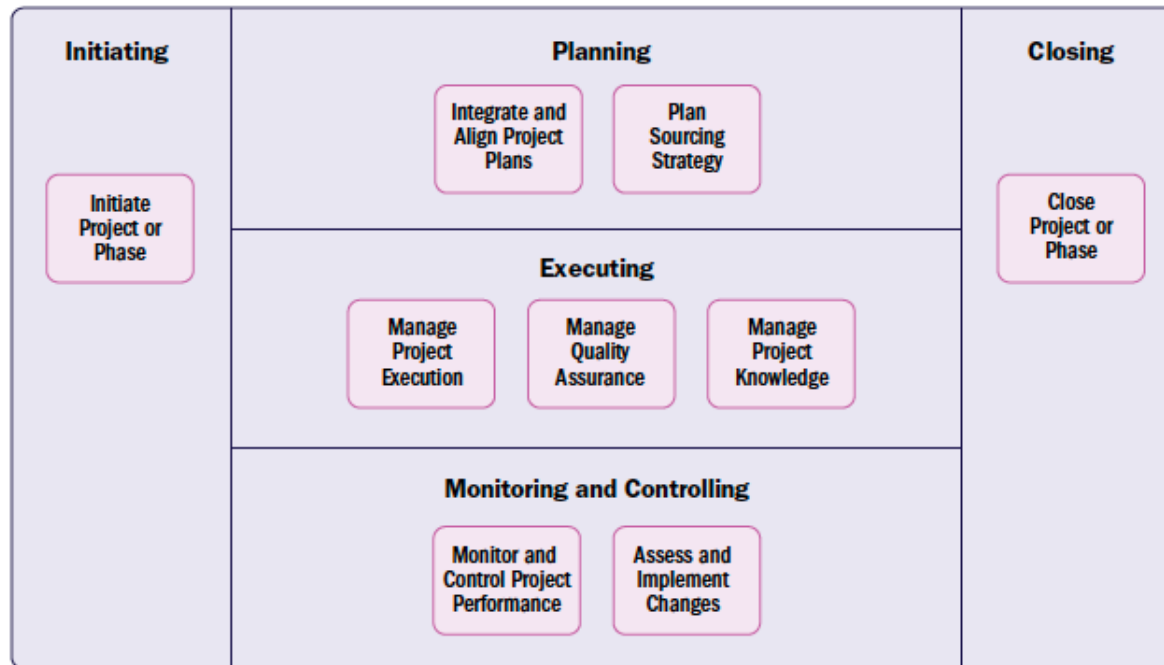


Figure 2-2. Governance Performance Domain Processes Overview

Source: PMBOK 8th Edition





Project Management

PMBOK Project Integration Management Processes

1. Develop **Project Charter**
2. Develop **Project Management Plan**
3. Direct and **Manage Project Work**
4. **Manage Project Knowledge**
5. **Monitor and Control Project Work**
6. Perform integrated **Change Control**
7. **Close Project or Phase**



Project Management

Prince 2 Project Integration Management Processes

1. Starting Up a Project (SU)

Establish project team and develop a project brief.

2. Directing a Project (DU)

Oversight of the project by the Project Executive and Project Board.

3. Initiating a Project (IP)

Develop a project initiation document.

4. Controlling a Stage (CS)

Management of work packages.

5. Managing Product Delivery (MP)

Delivery and acceptance of work packages.

6. Managing Stage Boundaries (SB)

Manage transition from one stage to other (Gateway process).

7. Closing a Project (CP)

Formal close-out of the project



Project Management

V-Model

- This is a project management method for development projects that originally comes from IT.
- In this case, the **waterfall model** is extended by test stages, with each development phase corresponding to test phases.



Project Management

Rational Unified Process (RUP)

- RUP includes a software development process on the one hand and corresponding tools from IBM on the other.
- It provides for iterations and defines the following four phases:
 - Inception
 - Elaboration
 - Construction
 - Transition



Project Management

Scrum

- Scrum is understood as a framework in the field of project management.
- It is a simple process model with few rules and is based on agile software development.



Our Approach and Focus Area



We will focus on the following:

1. PMBOK 6th & 8th Edition (most of the times)
2. Practical Approach as used in the industry
3. Focus on AI in Project Management rather AI Project Development.
4. Discussions in the class, in-person tutorials, & online consultations.

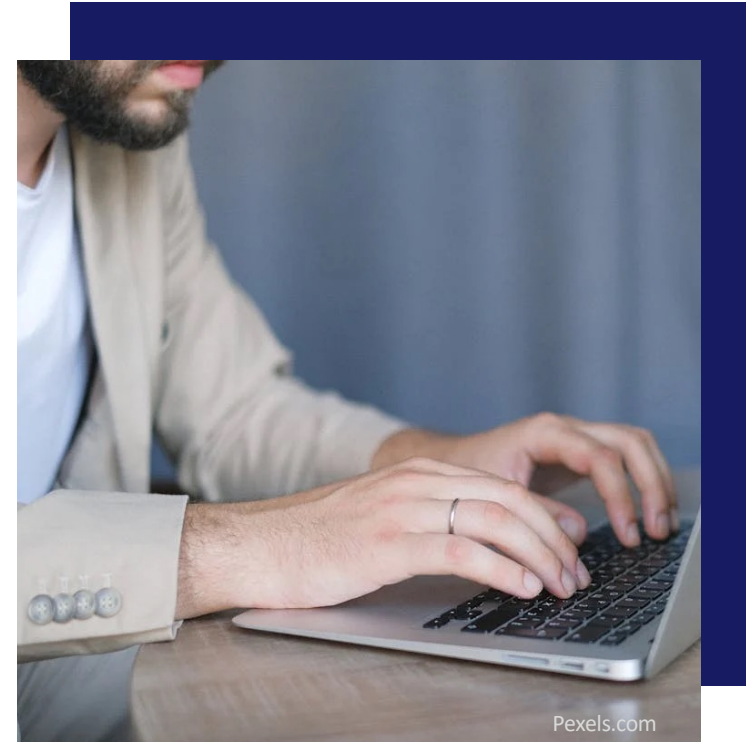


Project Manager

Person **leading the project team** achieving the project **objectives** within the **scope** boundaries.

Basic expectations

- **Scope**
- **Cost**
- **Schedule**
- **Quality**
- **Risk**





Project Manager

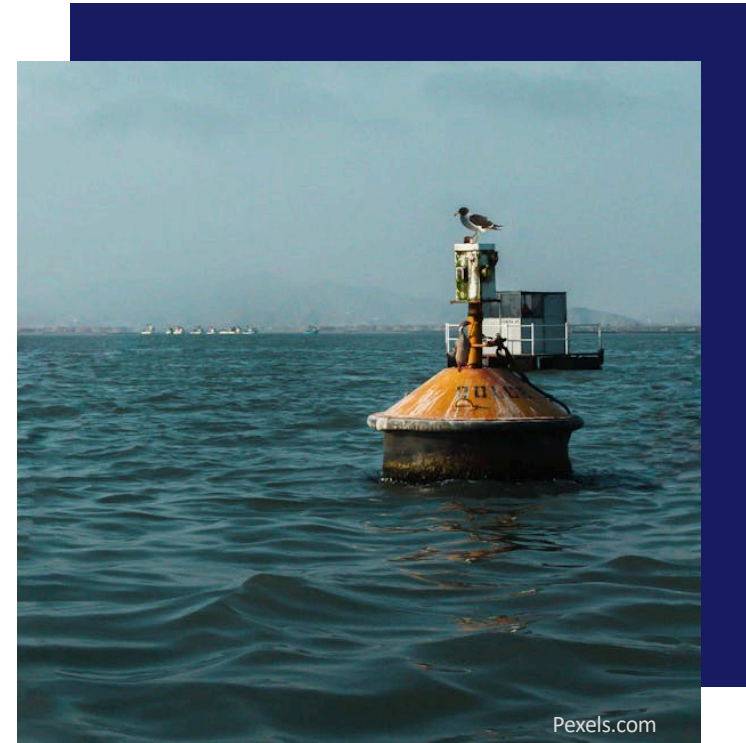
- Provide **leadership** for the project.
- Prepare a **Project Management Plan**.
- Manage **Resources, Risks, & Changes**.
- **Direct, coordinate, integrate, and motivate** the project team.
- **Communicate and engage** stakeholders.





Project Manager: Attributes

- Knowledge (**Business, Technical & Commercial**)
- **People management skills**, communication, negotiation, and social skills
- Able to **work under pressure** and stay focused to meet the objectives and deadlines
- **Inspire and motivate** others
- **Political and cultural** awareness





Project Team

- **Team reports** to the Project Manager.
- **Work Breakdown Structure (WBS)** is used to plan resources on a project.
- Project team members are **responsible to deliver the specific task** assigned to them.
- Team is selected to match the **right skills and experience** for the needs of the project.





Project Sponsor

- **Senior management personnel** from the owner side.
- Initiates the project through a **Project Charter** or any similar initiation document.
- **Authorizes the committal of resources** to the project.
- Oversees that the objectives of the project.



Project Stakeholders

- Project Sponsor/Customer
- End-users
- Project Team
- Consultants
- Contractors
- Suppliers and Vendors
- Functional Heads of Organization
- Project Management Office (PMO)
- Regulatory Agencies





Technology/ Software Development Projects

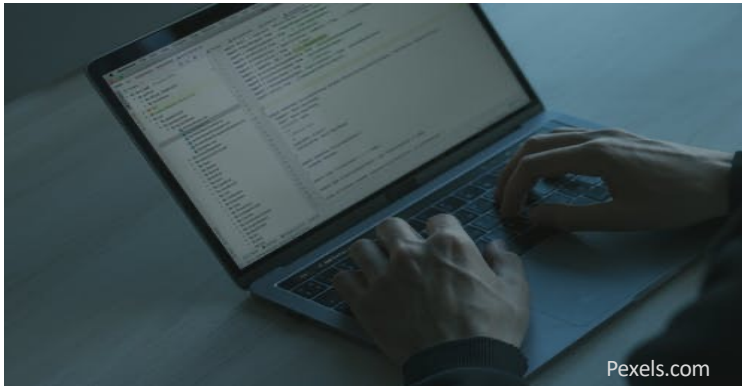
Engineer is lead

Examples:

- IT Software Development
- Data-center set up
- Technology development
- IT server room set up
- Others

Deliverables:

- Concept design
- Schematic design
- Detailed design and specifications
- Contract documents development.

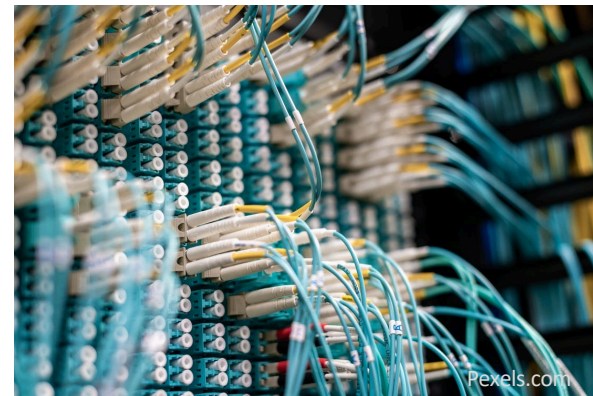




Technology/ Software Development Projects

Software
Development

Hardware
Set up

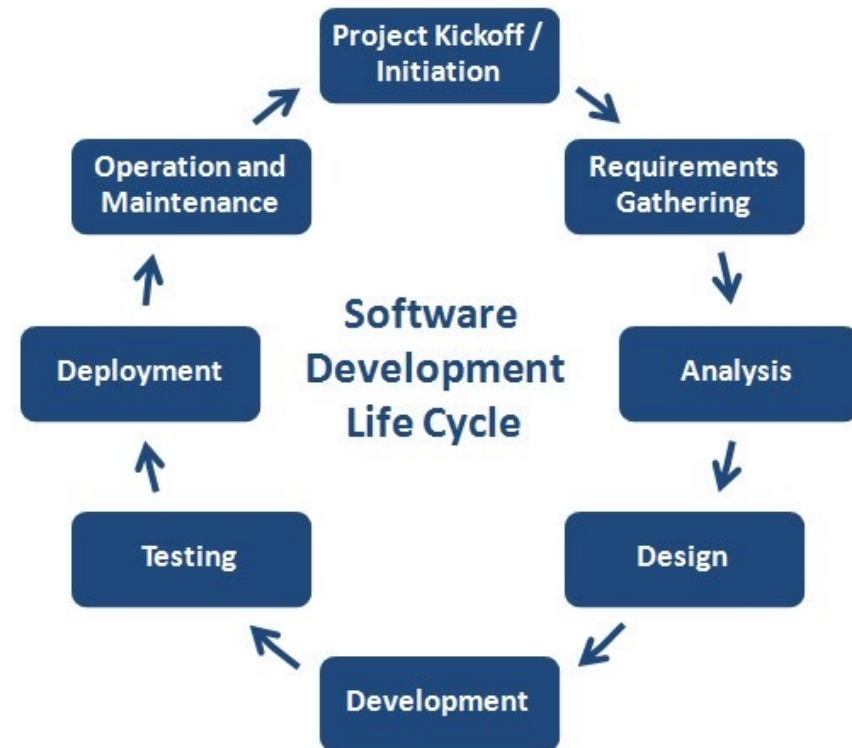




Software Development Life Cycle (SDLC)

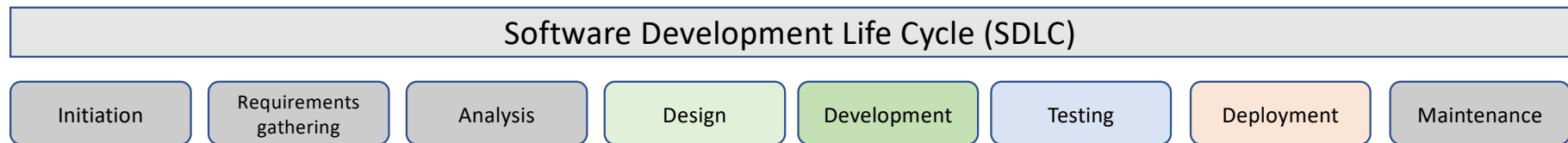
SDLC Phases:

1. Project Initiation
2. Requirements Gathering
3. Analysis
4. Design
5. Development
6. Testing
7. Deployment
8. Maintenance





Technology/ Software Development Projects

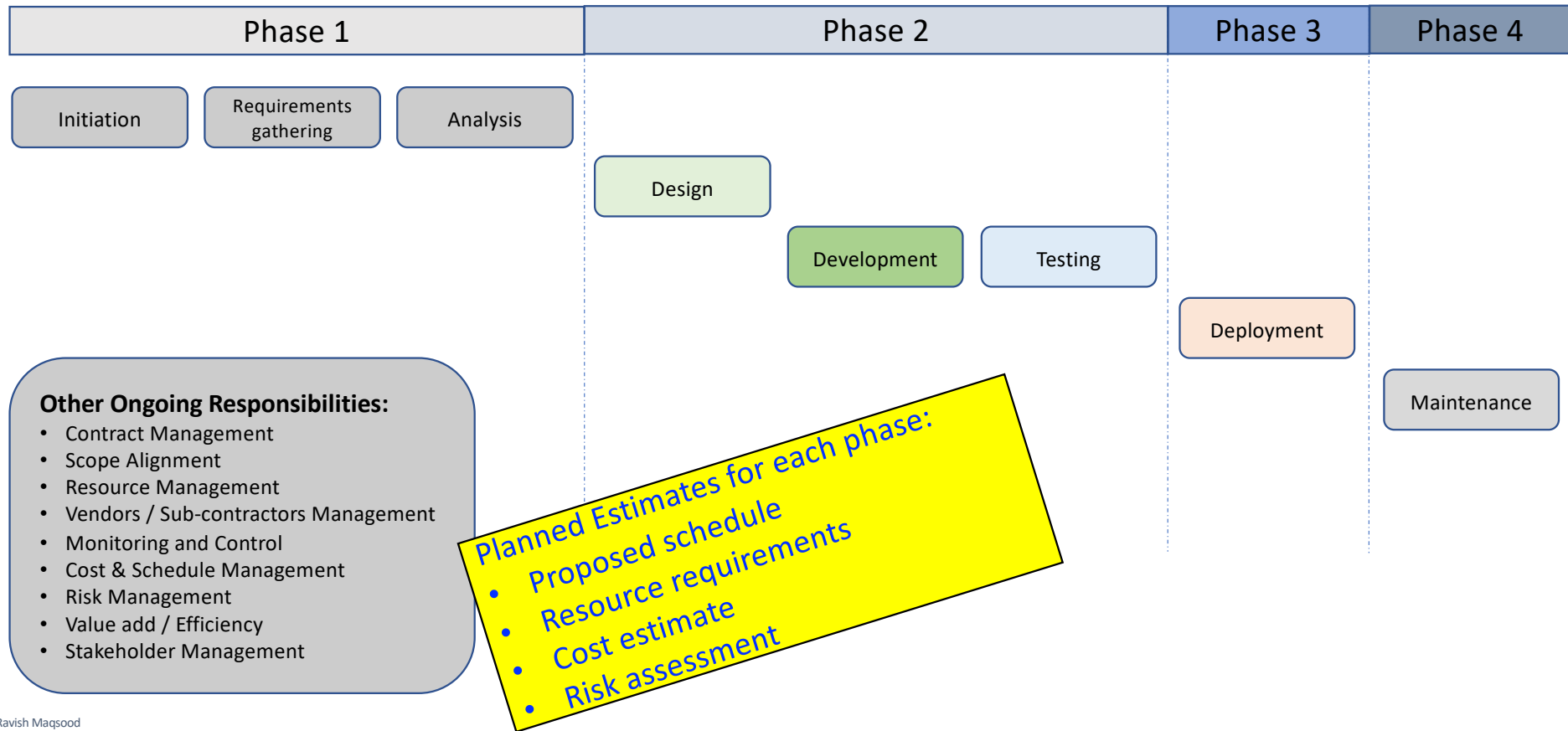


Project Manager could be from either of the disciplines:

- Information Technology
- Computer Science
- Data Science



Technology/ Software Development Projects





Software Development Industry

Computers

Internet

Programming

Software Development Industry Outlook

Market Size value in 2021 : ~430 bn

Revenue Forecast in 2028 : ~912 bn

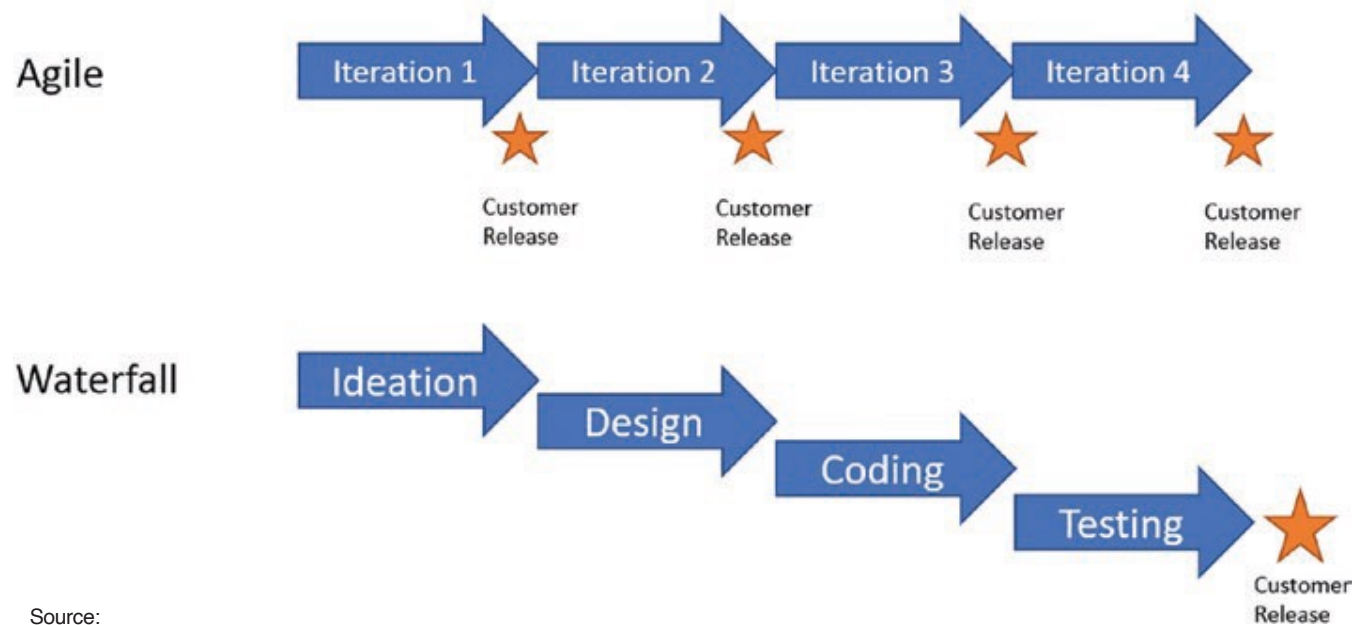
Growth rate (2021 – 2028) : CAGR of ~11.3%

CAGR : Compound Annual Growth Rate



Project Management Methodologies

Agile vs. Waterfall Lifecycle



Source:
AGILE PRODUCT AND PROJECT MANAGEMENT
A STEP-BY-STEP GUIDE TO BUILDING THE RIGHT PRODUCTS RIGHT
Mariya Breyter



AI Development Projects

Engineer is lead

Examples:

- AI solutions development & deployment.
- Business workflows automation
- Various industries (IT, Construction, Manufacturing, Aerospace, Warehouse, Logistics, etc.).
- Others.

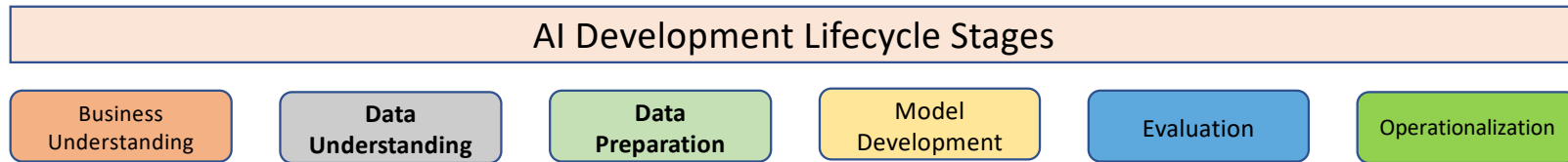
Deliverables:

- Workflow development and deployment.
- Evaluation.
- Operationalization.

Pexels.com



AI Development Projects



Project Manager could be from either of the disciplines:

- Computer Science
- Data Science
- Information Technology
- Other discipline with knowledge and interest in data



Artificial Intelligence (AI)

High level Perspective



Artificial Intelligence (AI): Definition (Simplified)

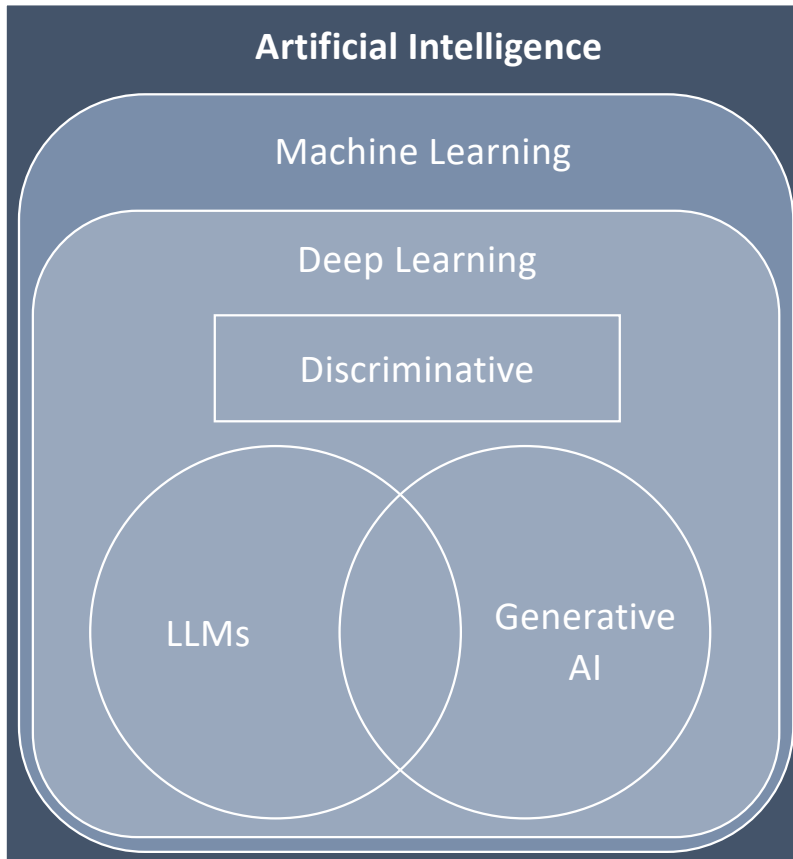
Artificial intelligence, or **AI**, is a technology that enables computers and machines to simulate human intelligence and problem-solving capabilities.

How does AI work?

In general, AI systems work by ingesting large amounts of labelled training data, analysing that data for correlations and patterns, and using these patterns to make predictions about future states.



Artificial Intelligence



- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models (LLMs)



AI in Project Management or AI Project?

High level Perspective, introduction



Artificial Intelligence and Project Management

Our focus in this course



AI in Project
Management



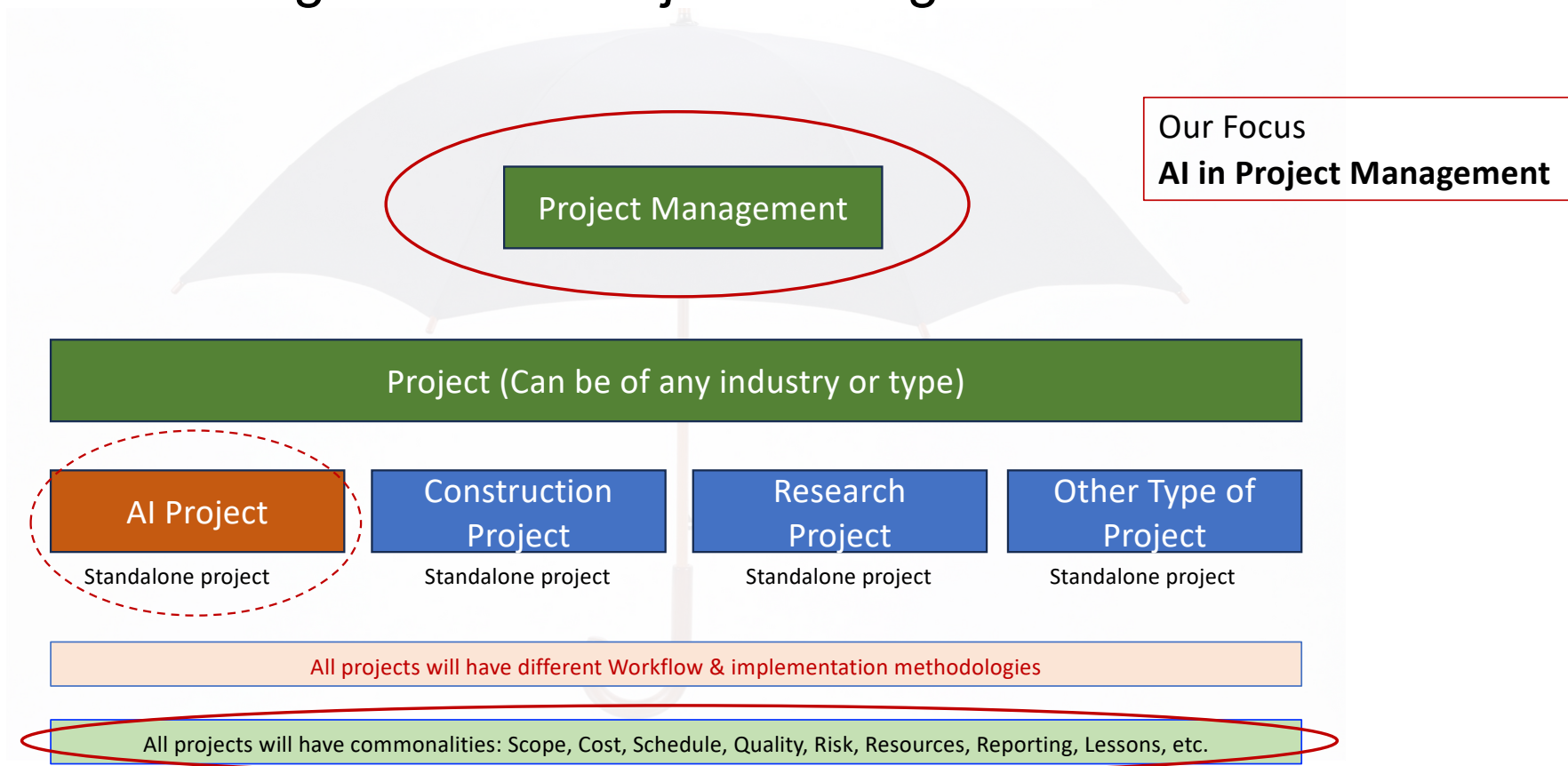
AI Project



Pexels.com



Artificial Intelligence and Project Management





AI in Project Management

High level Perspective, introduction



Artificial Intelligence and Project Management

- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models (LLMs)

How does
Artificial Intelligence
relate to
Project Management?



Pexels.com



Artificial Intelligence: Relation to Project Management

- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models (LLMs)

- **Machine Learning**

- Risk Management: Predict potential risks and issues based on historical data and current trends.
- Resource Optimization.
- Schedule tracking.
- Cost Management (Budget and Actual).
- Quality Control.



Artificial Intelligence: Relation to Project Management

- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models (LLMs)

- **Deep Learning**

- Risk Management.
- Natural Language Procession (NLP): Automated reporting from the data.
- Image and Video Analysis (Site Monitoring & Quality Control)- performance deviation.
- Resource matching (people and machinery).
- Fraud detection in financial reports.



Pexels.com

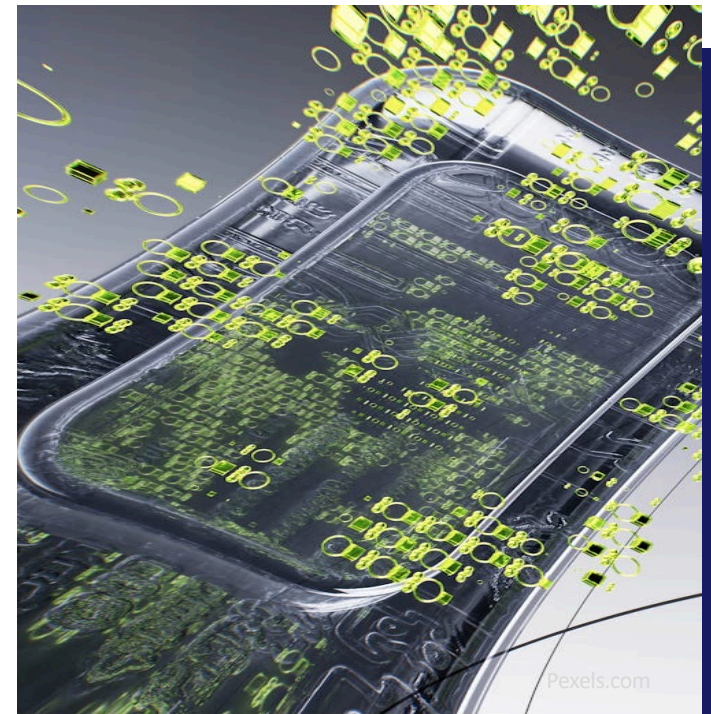


Artificial Intelligence: Relation to Project Management

- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models (LLMs)

- **Generative AI / LLMs**

- Automated Documentation and Reporting (Plans, Proposals).
- Emails, Meetings Agenda, and Meetings key pointers.
- Project scenarios planning and Simulation (different outcomes, resource allocation, etc.).
- Customized training materials





Artificial Intelligence (AI)

Why to use AI in project execution?



AI in Project Management: Is there a need?

- Out of projects across the industries, **12% of the projects fails** (PMI 2021).
- **35%** of the projects **over run costs** or **experience scope changes**.
- Indicates that **there is a huge potential to improve** project execution.
- Commonly identified causes by Najdawi and Shaheen, 2021:
 - Outdated or not relevant technologies used on the projects.
 - Ineffective project management methodologies.
 - Poor leadership.
 - Lack of commitment from stakeholders.
- **Can AI help?**



Market Data

Source: Statista



Machine Learning Market Size and Data

Source: Statista



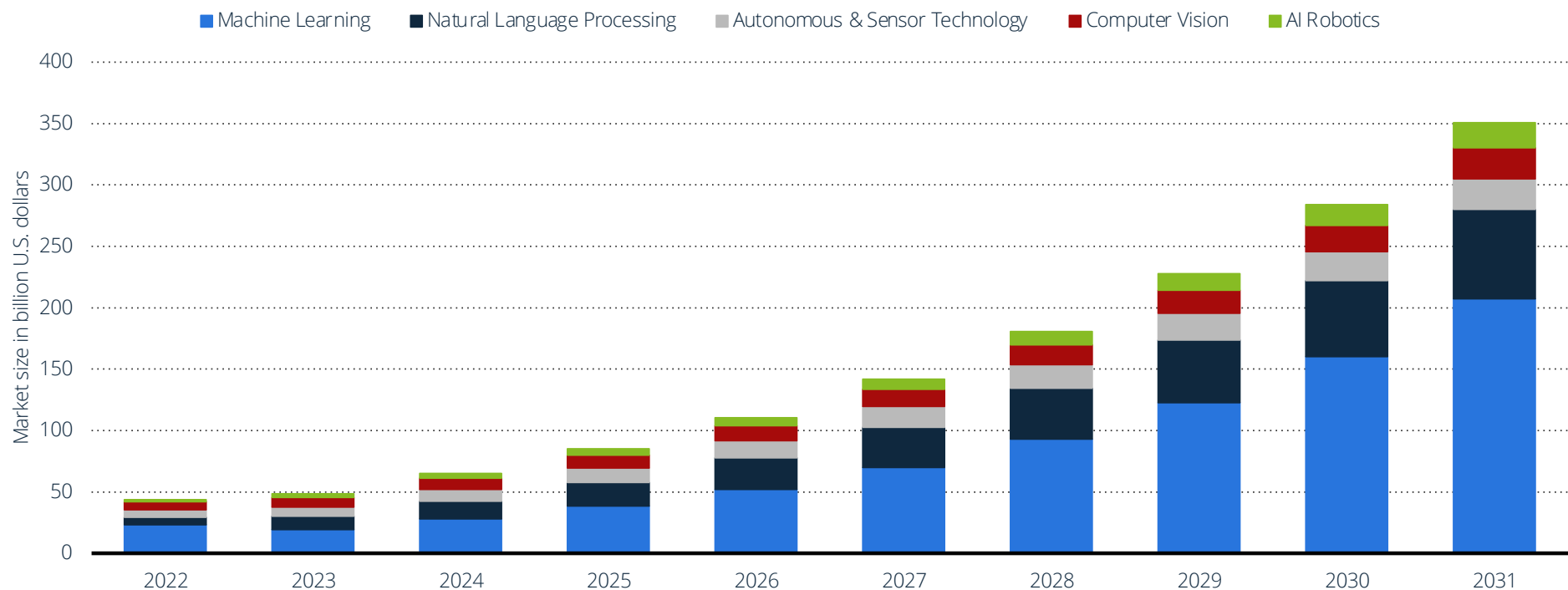
Artificial Intelligence (AI) Market in the Asia-Pacific (APAC):

Area (segment)	2025	2031 (forecast)
Machine Learning	38.45 billion USD	207.27 billion USD

statista 

Market size of the artificial intelligence (AI) market in the Asia-Pacific region from 2022 to 2031 (in billion U.S. dollars)

AI market size APAC 2022-2031, by segment



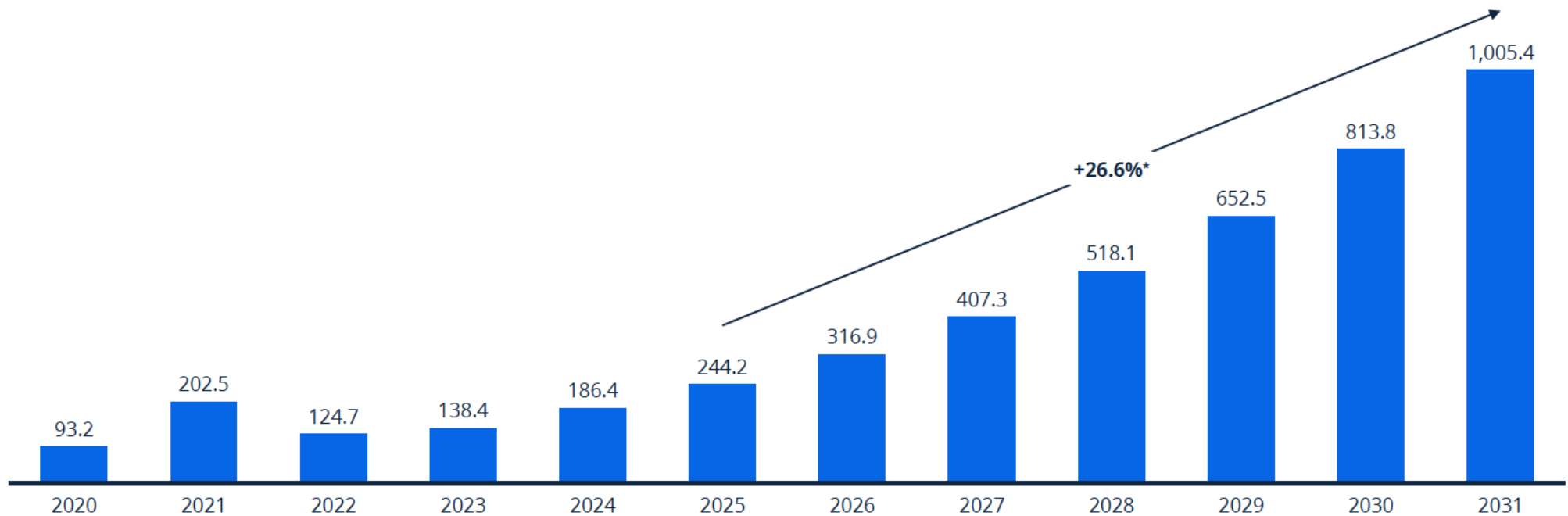
Note(s): APAC; 2022 to 2031

Prepared by: Ravish Madhukar
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Further information regarding this statistic can be found on [page 8](#).
Source(s): Statista Market Insights; Statista; ID 1622820

Global AI market is projected to exceed one trillion U.S. dollars by 2031

Global revenue projection (1/6)

Global artificial intelligence market, in billion U.S. dollars



20 Notes: *CAGR: Compound Annual Growth Rate

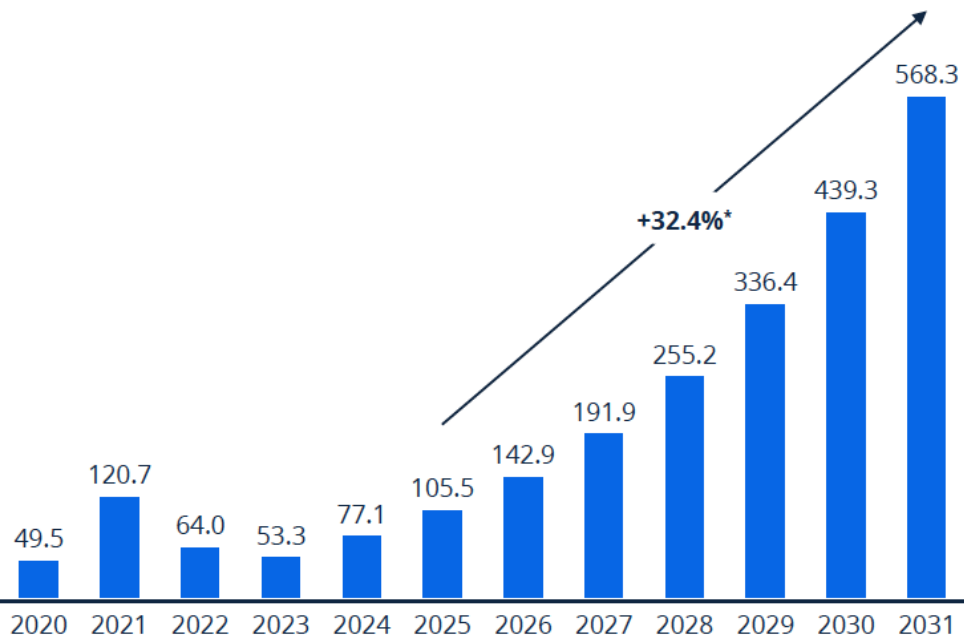
Sources: Statista Market Insights

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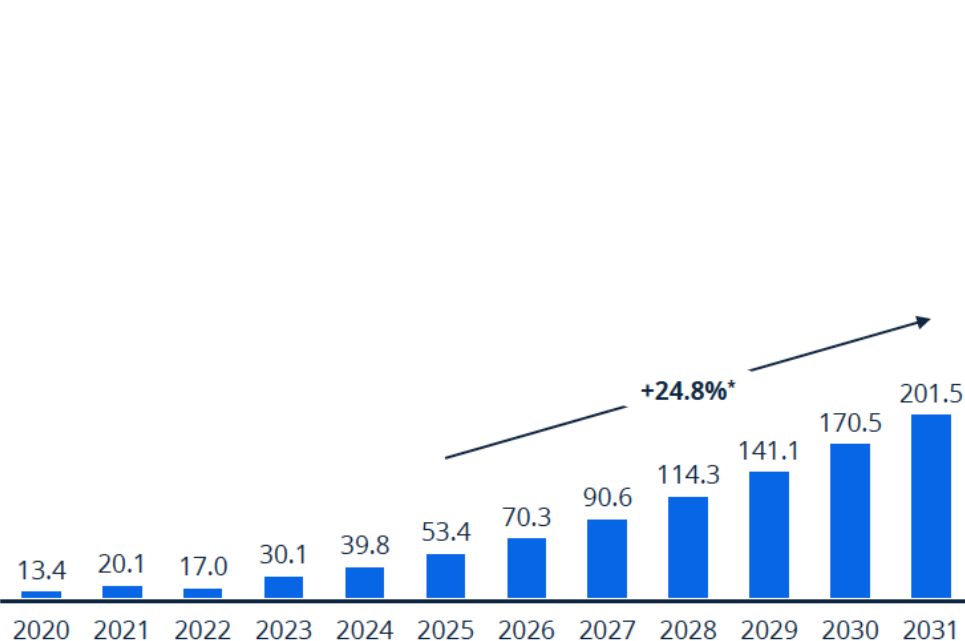
Machine Learning market projected to surpass 500 billion U.S. dollars by 2031

Global revenue projection (2/6)

Global machine learning market, in billion U.S. dollars



Global natural language processing market, in billion U.S. dollars



21 | Notes: *CAGR: Compound Annual Growth Rate

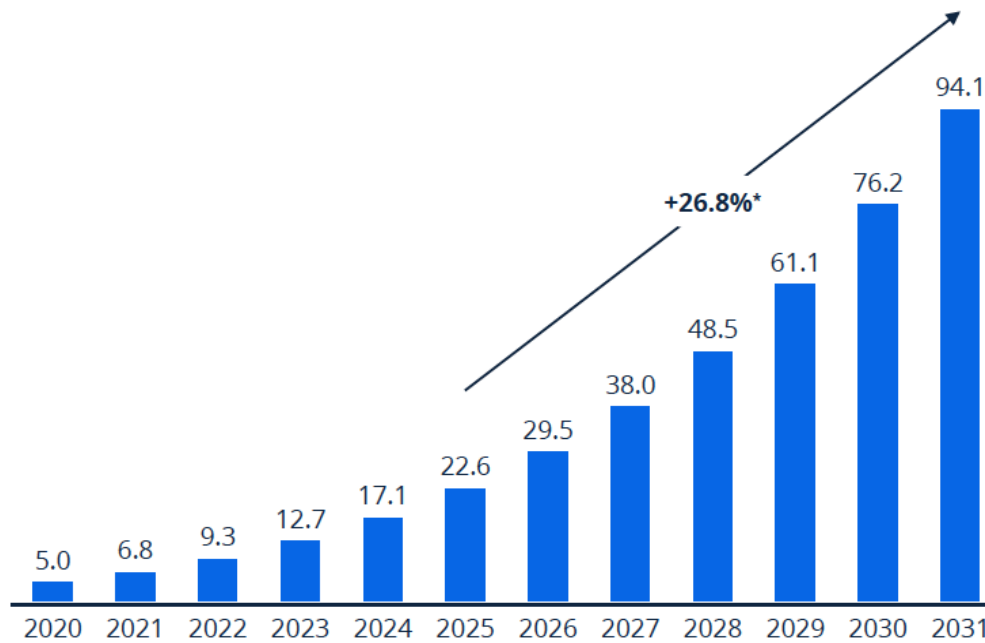
Sources: Statista Market Insights

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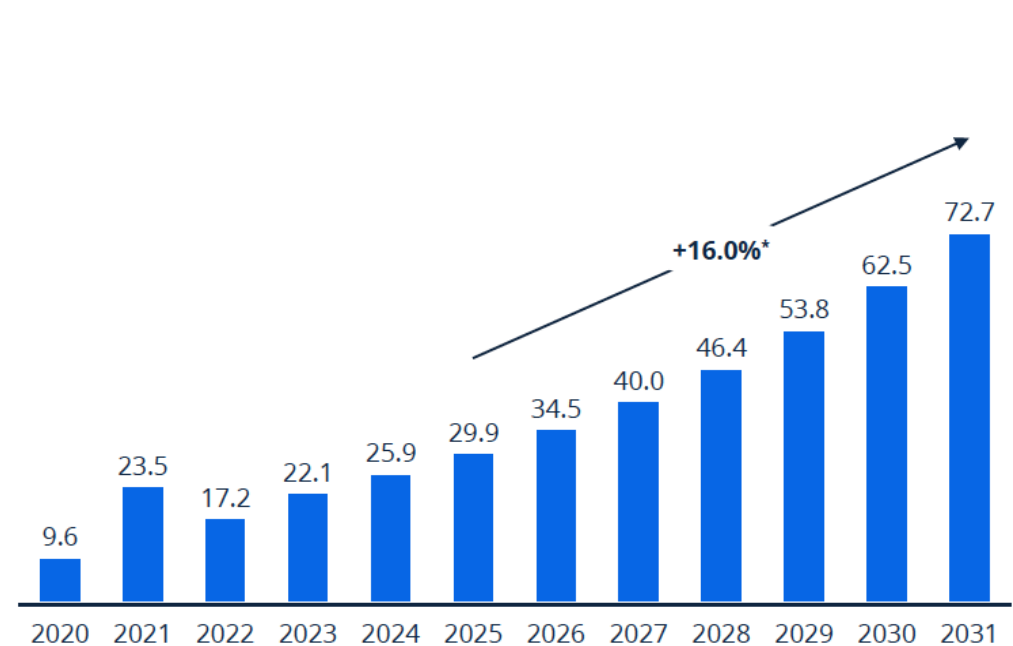
Global AI robotics market projected to reach nearly 95 billion U.S. dollars

Global revenue projection (3/6)

Global AI robotics market, in billion U.S. dollars



Global computer vision market in billion, U.S. dollars



22 | Notes: *CAGR: Compound Annual Growth Rate

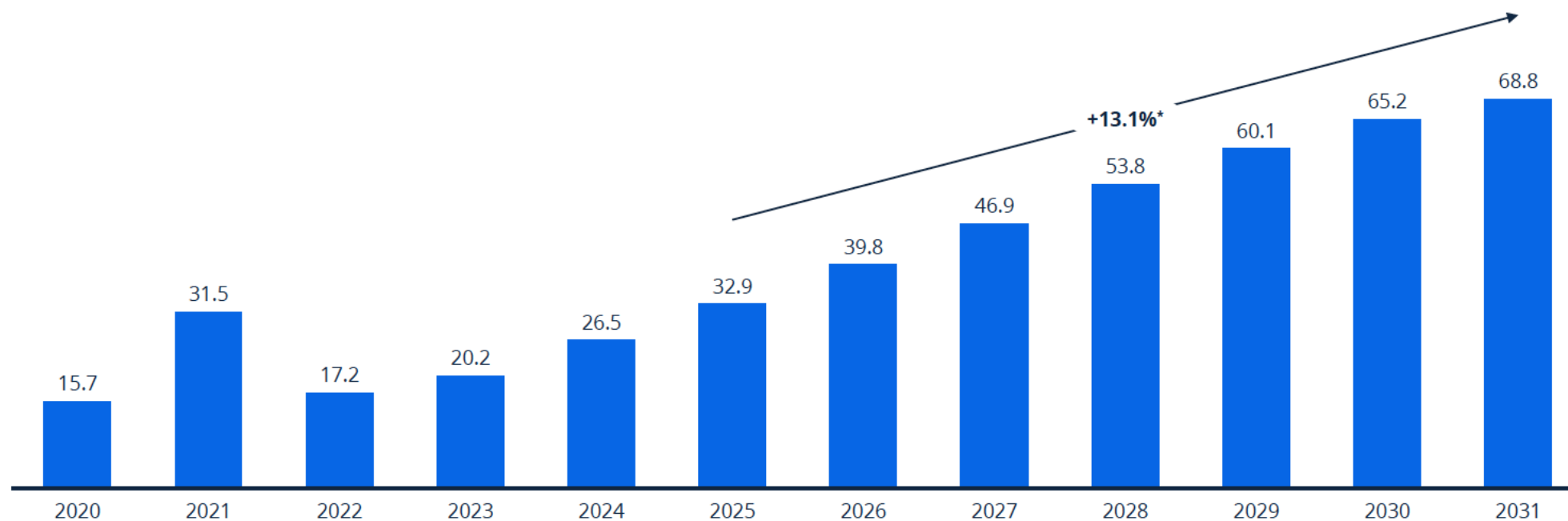
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Autonomous & sensor tech to be worth 68.8 billion U.S. dollars by 2031

Global revenue projection (4/6)

Global autonomous and sensor technology market, in billion U.S. dollars



23 | Notes: *CAGR: Compound Annual Growth Rate

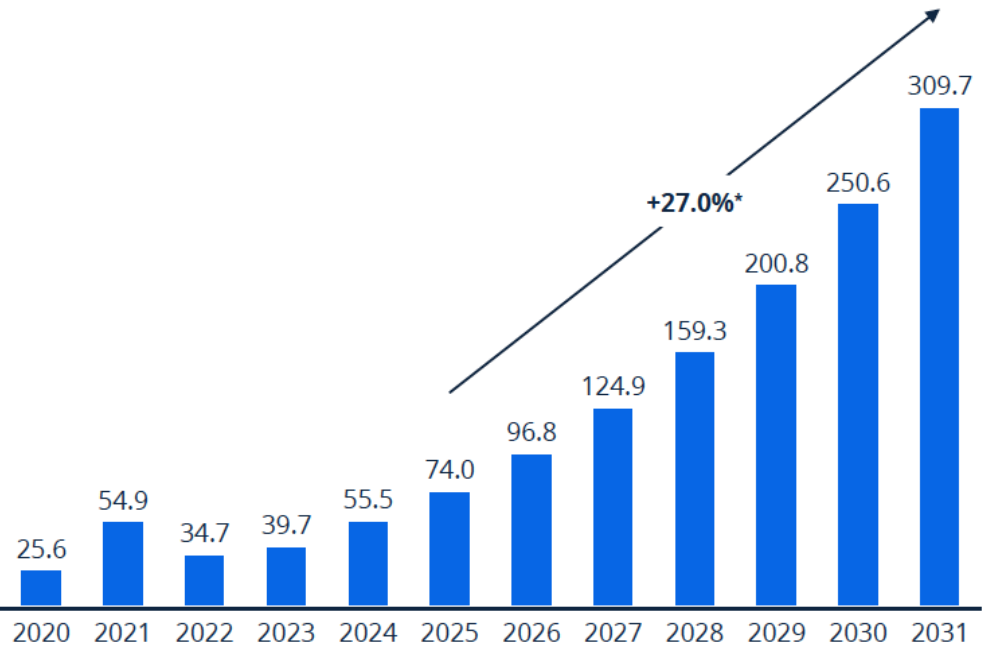
Sources: Statista Market Insights

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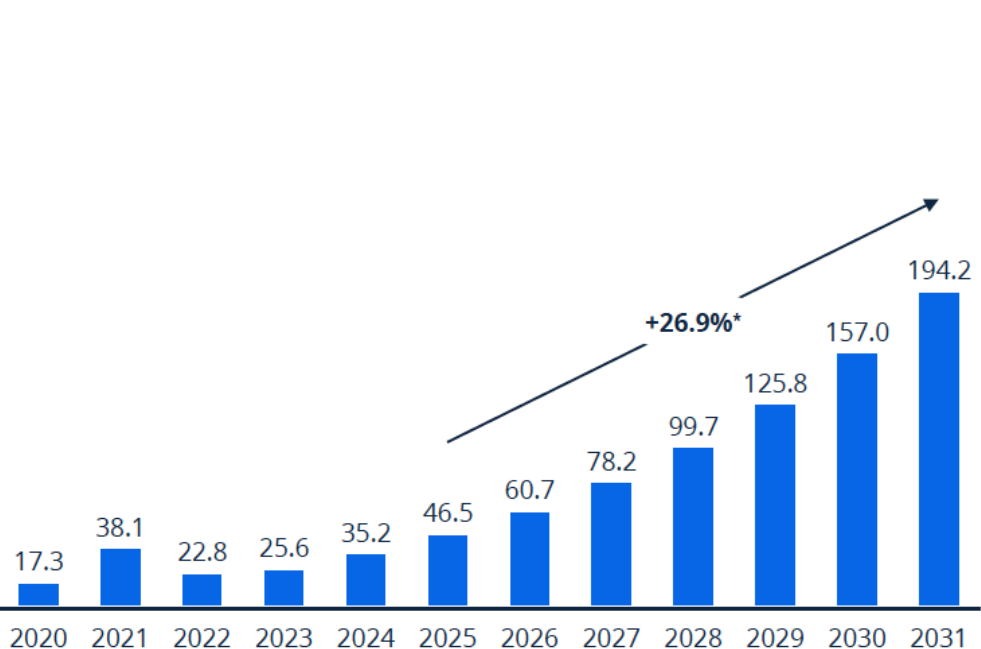
China's AI market projected to be valued at 195 billion U.S. dollars by 2031

Global revenue projection (5/6)

U.S. artificial intelligence market, in billion U.S. dollars



China artificial intelligence market, in billion U.S. dollars



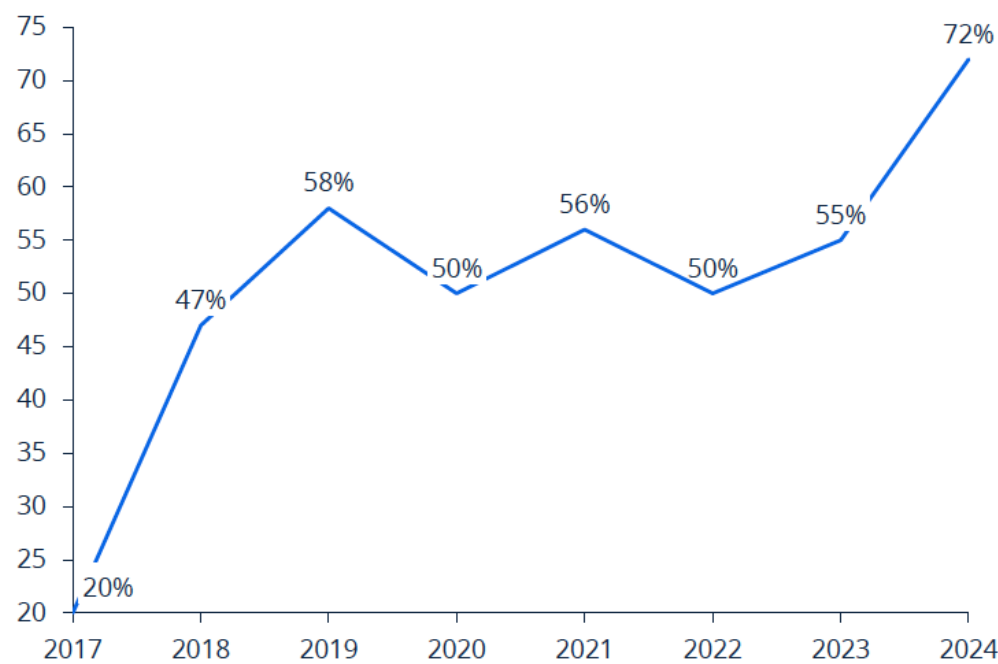
24 | Notes: *CAGR: Compound Annual Growth Rate
Sources: Statista Market Insights

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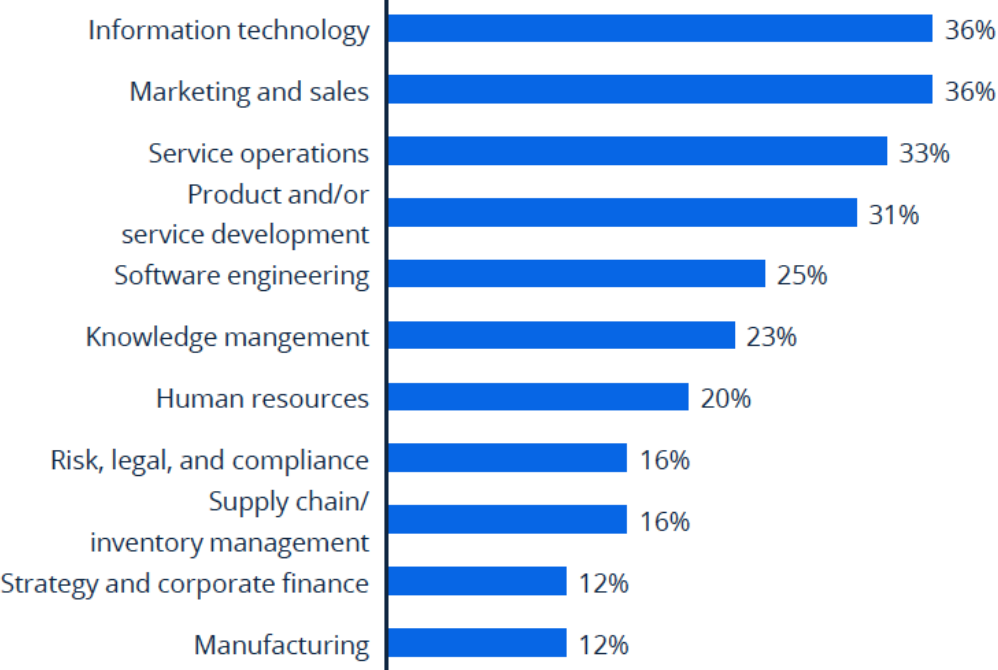
Global AI adoption rate in businesses stood at 72 percent in 2024

AI adoption (1/4)

Global AI adoption rate in businesses



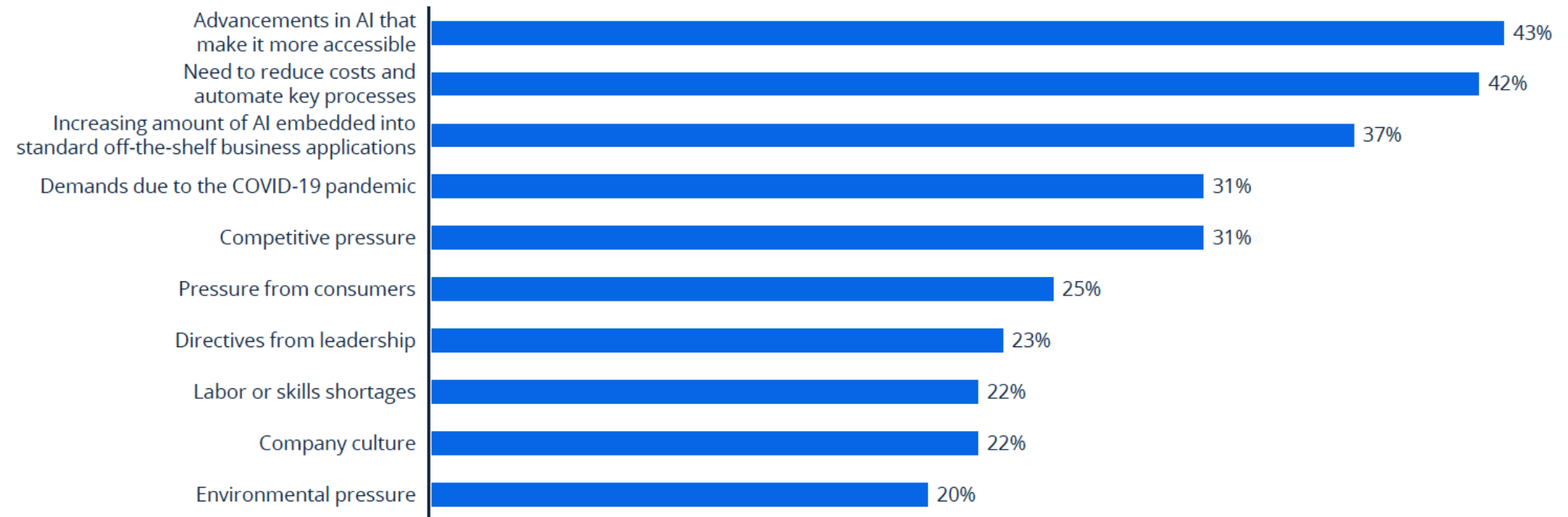
Use of AI by business function in 2024



Advancements that make AI more accessible is the top factor driving adoption

AI adoption (2/4)

The top factors driving AI adoption



29 Notes: as of 2024

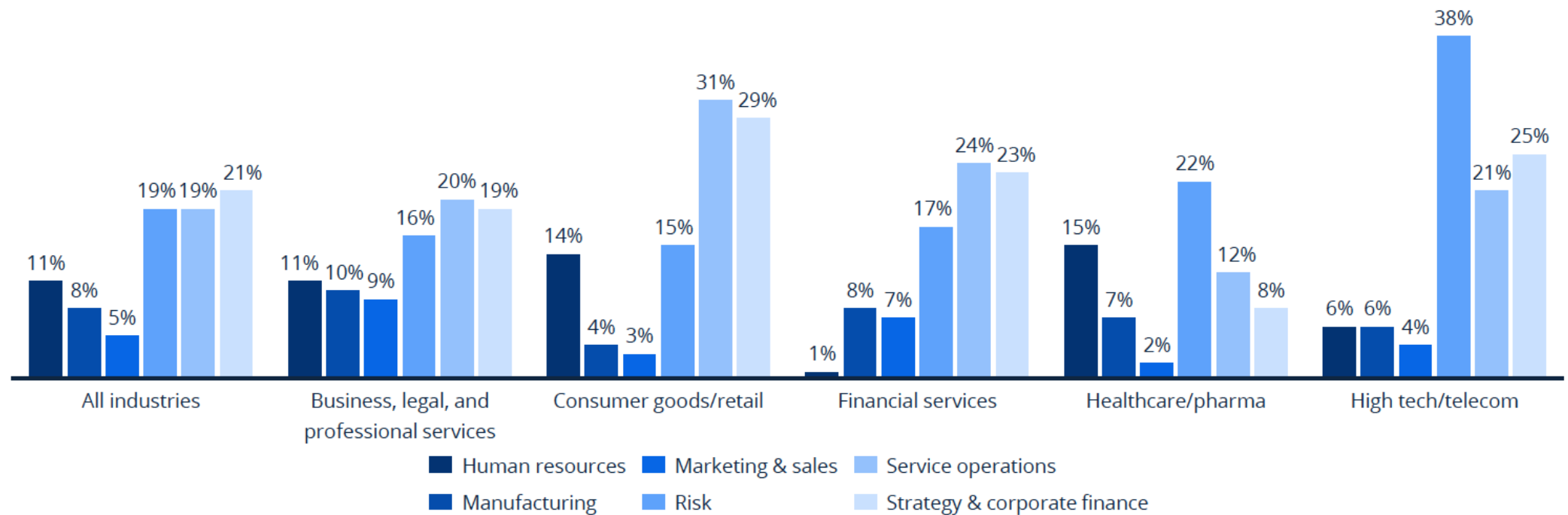
Sources: IBM Global

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AI adoption rates are growing across industries

AI adoption (3/4)

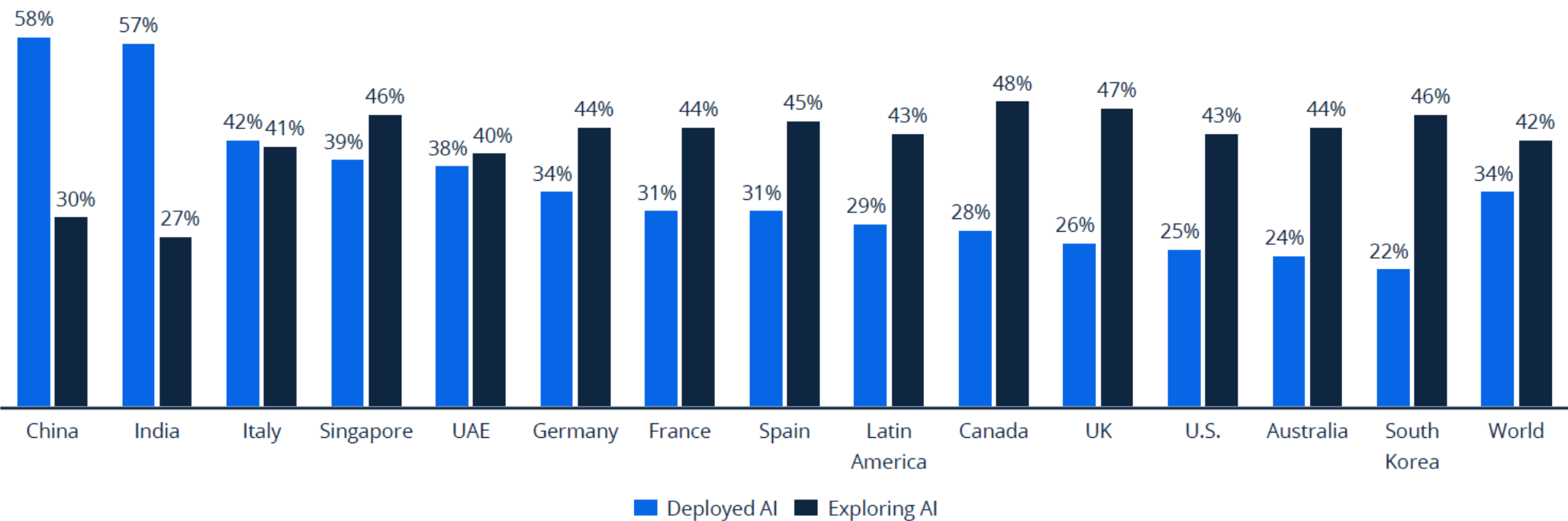
AI adoption rate by industry in 2024



China and India are among the leading countries in AI deployment

AI adoption (4/4)

AI adoption rates around the world in 2024





CAGR of AI Spending in APAC

Source: Statista

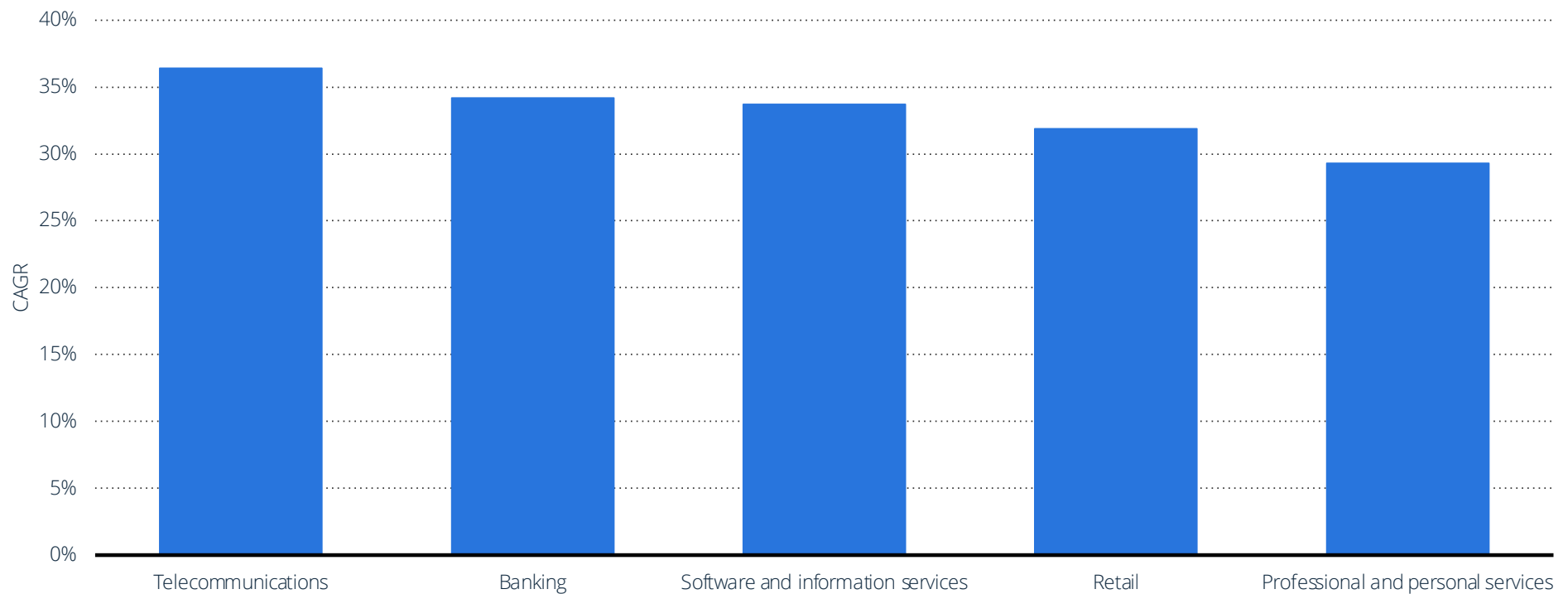


Investment in AI and GenAI

In the **Asia-Pacific region**, the software and information services industry leads in absolute amount of investment on AI and GenAI in 2025, at around **36.8 billion** U.S. dollars.

Compound annual growth rate (CAGR) of AI spending in the Asia-Pacific region from 2023 to 2028, by selected industry

CAGR of AI spending APAC 2023-2028, by industry



Note(s): APAC; 2023 to 2028

Further information regarding this statistic can be found on [page 8](#).

Source(s): IDC; [ID 1620295](#)

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Digital Transformation Market Size and Data

Source: Statista



Digital Transformation

Digital transformation refers to the adoption of digital technology to transform business processes and services from non-digital to digital.

This encompasses, among others, moving data to the cloud, using technological devices and tools for communication and collaboration , as well as automating processes.



Digital Transformation

What is driving digital transformation?

- COVID-19 pandemic increased the digital transformation.
- Customer demand and the need to be on par with competitors.
- Responding to changing markets and enhance innovation.

worldwide

Digital transformation spending worldwide 2017-2027



Note(s): Worldwide; 2017 to 2024

Further information regarding this statistic can be found on [page 8](#).

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Software Market Data

Source: Statista



Enterprise Software

What is enterprise software?

- Enterprise software is the software primarily used to satisfy various business needs.
- It helps companies meet their goals by boosting enterprise productivity and efficiency – for example through helping to track and improve supply chains, interact with customers, or manage resources.
- There are a wide range of applications under the term “enterprise software”, with accounting software, office suites, human resource management (HRM) systems, and enterprise resource planning (ERP) being a few of them.



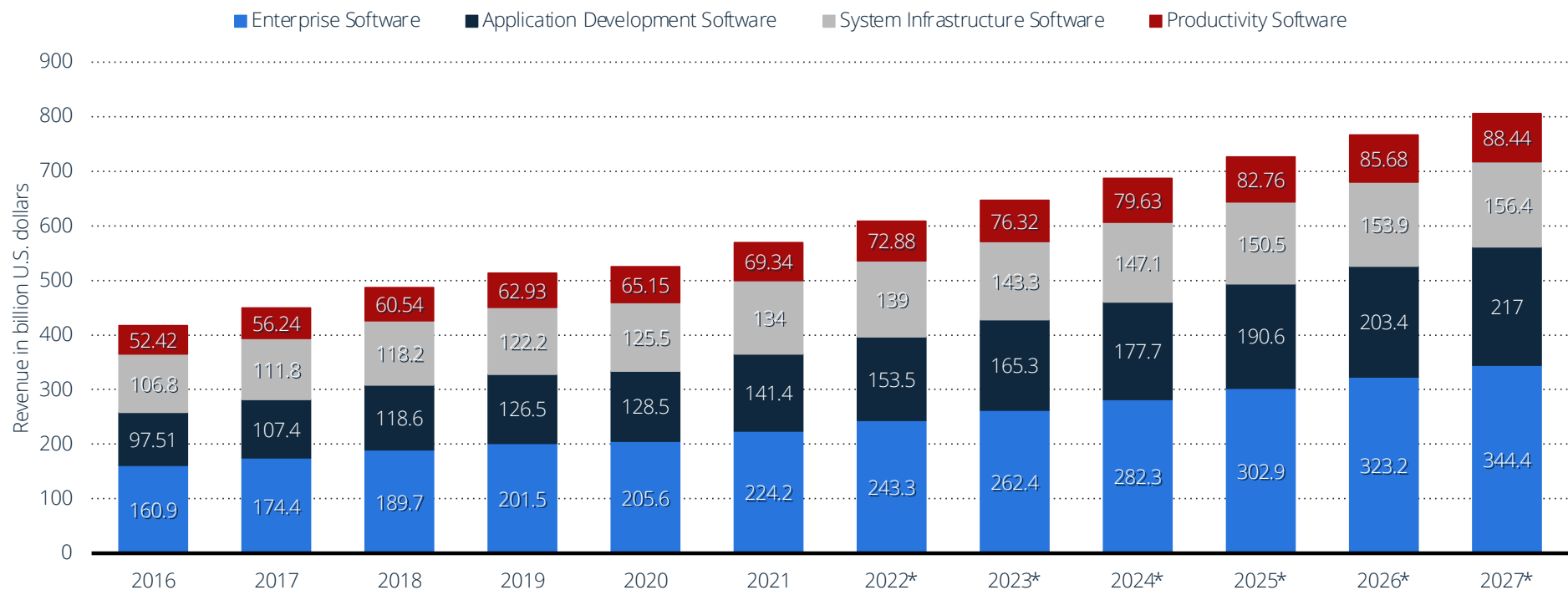
Enterprise Software

How do most global enterprises deploy their ERP solutions?

- Enterprise resource planning (ERP) is a type of software organizations used to manage day-to-day business activities like supply chain operations, accounting, or project management, among others.
- These software solutions can be deployed through different models.
- Most companies worldwide report on-premise solutions as their primary method for ERP deployment followed by cloud ERP systems and software-as-a-service solutions.
- Over 30 percent of companies worldwide reported cost overruns in their ERP implementation projects as of 2021.

Revenue of the software market worldwide from 2016 to 2027, by segment (in billion U.S. dollars)

Software market revenue worldwide 2016-2027, by segment



Note(s): Worldwide; 2016 to 2021

Further information regarding this statistic can be found on [page 8](#).

Source(s): Statista; Statista Technology Market Insights; ID 954176
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Where AI can be used in Projects?

AI in Project Management



AI tools in Project Management: Where to use? (non-exhaustive)

- Project Management Plan
- Scope development
- Planning and scheduling
- Budget development
- Cost management
- Progress monitoring
- Resource allocation & management
- Risk assessment & management
- Procurement Management
- Document control & management
- Change management
- Project Communication
- Change management
- Vendors/ Suppliers/ evaluation
- Sub-contractor evaluation
- Progress payments
- Financial background checks
- Project specifications development
- Quality management
- Inspections
- Trainings
- Standard Operating Procedures (SOPs) development.



Artificial Intelligence: Relation to Project Management

- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models (LLMs)

• Design Phase

- Idea Generation and Brainstorming.
- Requirement Analysis and Gathering Data: Natural Language Processing (NLP) to analyze client requirements from documents to extract critical needs.
- Design Optimization: generate multiple options based on given criteria.
- Simulation and Testing: Simulate various design scenarios.
- Prototypes, 3D model generation, Virtual and Augmented Reality models.
- Design Validation and Verification against specifications, standards, regulations, best practices, etc.
- Budget estimation.
- Many more.....



Artificial Intelligence: Relation to Project Management

- Machine Learning
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• Procurement Phase

- Supplier Selection and Evaluation: Based on historical data on supplier performance, unit rates comparison, and market intelligence.
- Inventory Optimization: predicting demand fluctuations, reducing the risk of overstocking or stockouts.
- Automated Procurement Processes: Purchase orders and sub-contracts drafting.
- Suppliers and Sub-Contractors Risk Management / Fraud Detection: monitors financial instability, compliance issues, and geo-political issues to raise early flags.
- Many more.....



Artificial Intelligence: Relation to Project Management

- Machine Learning
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• Execution Phase

- Automated Planning and Scheduling.
- Real-Time Tracking.
- Automated Reporting.
- Process Optimization through identification of deficiencies.
- Defect Detection (Quality issues).
- Predictive Analysis.
- Resource Optimization.
- Budget and Cost Tracking.
- Many more.....



Artificial Intelligence: Relation to Project Management

- Machine Learning
- Deep Learning
- Generative AI
- Large Language Models (LLMs)

What else can you think of?



Summary

- This is a unique topic helpful to understand the project execution methodologies.
- Recorded lectures, in-person tutorials, and class exercises shall be used to deliver this course. Class attendance and participation is marked.
- Different standards and approach available to project management.
- PMBOK 6th edition is commonly used as a reference in the industry.
- Our focus will be AI usage in project management for various industries projects.



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End of class



Let the journey begin!

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